

M13 Batch 9 — Full Set

M13.8 Instruments (ATA 31) · PDF pages 768–917

Remote compass systems, glass cockpit and EFIS displays, central warning systems and the ground proximity warning system.

417 questions · Level 3: 151 · Level 2: 187 · Level 1: 79

● marks a question built on a fact your instructor highlighted or underlined in the manual (370 of 417). Topics run in manual page order; inside each topic the questions are grouped by level, and the answers with explanations follow at the end of that topic.

No calculations, no unit conversions, no worked examples, nothing type-specific unless it was marked.

Remote compass systems and slaving

PDF pages 771–787 · 73 questions

LEVEL 1 — RECALL

1. Where on an aeroplane are the flux valves located ●
 - a. at the wing tips, in the vertical stabiliser or in the tail cone
 - b. in the nose radome, in the forward cargo bay or in the wheel well
 - c. under the cockpit floor, in the engine pylons or in the centre tank
2. Where is the flux valve of a helicopter usually installed ●
 - a. on the roof of the cabin, in the rotor head or on the upper deck of the engine
 - b. on the bottom of the tail boom, the vertical stabilizers or in the vertical fin
 - c. under the cockpit floor, in the nose compartment or on the landing gear skid
3. A directional gyro is left running for a long flight leg. Which disadvantage shows up? ●
 - a. a falsified reading in the region of the magnetic poles
 - b. a falsified heading during accelerations and turns
 - c. the drift
4. Which disadvantage do magnet compasses and flux valves have ●
 - a. a heading which lags behind by a fixed amount in the climb
 - b. a falsified heading during accelerations and turns
 - c. a drift which grows with the running time of the unit
5. What do the two acronyms in the name ADF RMI stand for ●
 - a. Automatic Direction Finder and Radio Magnetic Indicator
 - b. Airborne Direction Finder and Radio Magnetic Instrument
 - c. Automatic Deviation Finder and Remote Magnetic Indicator
6. What do the arrows in the ADF RMI point to ●
 - a. the magnetic north found by the flux valve
 - b. the course selected on the compass rose
 - c. tuned broadcast stations
7. Which three units does a remote compass with ADF/RMI have ●
 - a. a flux valve, a compass coupler and an ADF/RMI
 - b. a directional gyro, a compass coupler and a compass controller
 - c. a flux valve, a directional gyro and an ADF/RMI

8. Where are the amplifiers that raise the voltage of weak signals installed ●
 - a. in the automatic direction finder indicator
 - b. in the integrated instrument system rack
 - c. in the housing of the directional gyro
9. Which errors are adjusted by the screws on the flux valve compensation device ●
 - a. the A error only
 - b. the B and C errors
 - c. the A and B errors
10. How is the A error of a remote compass system compensated ●
 - a. by turning the adjustment screws of the compensation device
 - b. by turning the flux valve
 - c. by turning the N – S potentiometer of the remote compensator
11. Which units make up a remote compass system with a compass coupler ●
 - a. flux valve, DG, compass controller and compass coupler
 - b. flux valve, DG, ADF RMI and compass coupler
 - c. flux valve, DG, compass controller and remote compensator
12. With which control can any desired heading be set ●
 - a. the SET HDG knob
 - b. the DG – SLAVED switch
 - c. the synchronizing knob
13. Fast synchronisation is unserviceable on a compass system. Which unit holds the control logic? ●
 - a. in the compass coupler
 - b. in the directional gyro
 - c. in the compass controller
14. To which value does fast synchronisation raise the update rate of the magnetic heading ●
 - a. 70 °/min.
 - b. 700 °/min.
 - c. 7 °/min.

LEVEL 2 — UNDERSTAND AND EXPLAIN

15. A flux valve is to go at a wing tip rather than near the cabin. Why is that place chosen? ●
 - a. to keep them inside the pressurised area so they can be reached in flight
 - b. to exclude the influence of magnetic interference fields as far as possible
 - c. to bring them as close as possible to the heading indicators they supply
16. A helicopter flux valve is to be relocated and the shop asks what rules the choice of place. What governs it? ●
 - a. it must be far away from all magnetic fields of the helicopter
 - b. it must be close to the gearbox of the helicopter for a rigid mount
 - c. it must be inside the cabin of the helicopter to keep the run short
17. Why is the heading of a flux valve falsified while the aircraft accelerates ●
 - a. the valve swings so that it is also affected by the vertical magnetic field
 - b. the valve output is damped so that it follows the change far too slowly
 - c. the valve loses part of its excitation while the aircraft is accelerating

18. Which two units are combined to build a remote compass system that shows the correct heading ●
 - a. a directional gyro and a magnet compass
 - b. a flux valve and a directional gyro
 - c. a flux valve and a magnet compass
19. From which unit does the indicator of a remote compass system receive the stable heading information ●
 - a. from the magnet compass
 - b. from the directional gyro
 - c. from the flux valve
20. How is the difference between the magnetic field direction and the indicated heading used ●
 - a. it drives the compass rose of the indicator directly
 - b. it changes the excitation voltage of the flux valve
 - c. it adjusts the DG through a torque motor
21. The slaving torque motor applies a torque to the gyro. What does this produce in it? ●
 - a. a rigidity which locks the DG to magnetic north
 - b. a gyro precession which changes the DG direction
 - c. a friction brake which holds the DG direction
22. A crew has one ADF RMI on the panel. Which two kinds of information does it give them? ●
 - a. the magnetic heading and the distance to the station
 - b. a radio navigation indication and the magnetic heading
 - c. a radio navigation indication and the aircraft attitude
23. What does the slaving annunciator show the pilot ●
 - a. the difference between the selected course and the indicated heading
 - b. the difference between the indicated heading and the flux valve signal
 - c. the difference between the excitation voltage and its nominal value
24. Where is the difference between the indicated heading and the flux valve signal detected in this architecture
 - a. inside the flux valve
 - b. inside the compass coupler
 - c. inside the ADF RMI
25. A pilot reaches for the synchronizing knob on the ADF RMI. What can he change with it? ●
 - a. the mode in which the system operates
 - b. the indication of the compass rose
 - c. the rate at which the torque motor runs
26. What is protected by the fuse fitted in the integrated instrument system rack ●
 - a. the transformer which supplies the flux valve excitation
 - b. the amplifier which raises the compass rose motor voltage
 - c. the torque motor which is fitted inside the directional gyro
27. In which form does the detected field direction reach the control transformer in the indicator ●
 - a. as a single-phase signal
 - b. as a direct current signal
 - c. as a three-phase signal

28. A slaving control transformer is fitted in an ADF/RMI. What is its rotor connected to? ●
- the synchronisation knob
 - the compass card
 - the slaving torque motor
29. The north found by the flux valve matches the position of the compass rose. What appears at the slaving CT output? ●
- a steady direct voltage
 - no signal at all
 - a signal of maximum size
30. In which condition of flight does the information from the flux valve contain no errors? ●
- when flying a steady turn at a constant bank angle
 - when flying straight ahead without acceleration
 - when flying at a constant height in any manoeuvre
31. Which precision of heading indication is possible all the time with the remote compass system ●
- +/- 2°
 - +/- 5°
 - +/- 0.5°
32. Where is the amplified error signal from the control transformer sent ●
- to the compass rose motor
 - to the slaving torque motor
 - to the flux valve signal coils
33. After switch on the gyro is not aligned with magnetic north. How long can the automatic alignment take? ●
- up to 9 minutes
 - up to 90 minutes
 - up to 90 seconds
34. The annunciator needle of the ADF/RMI sits at its central position. What does this mean? ●
- the slaving amplifier has failed
 - the system is in the free azimuth mode
 - the slaving CD is zeroed out
35. What does the annunciator on the ADF/RMI display ●
- the excitation voltage supplied to the flux valve
 - the output voltage of the slaving torque motor
 - the output voltage of the slaving amplifier
36. Between which two units can a remote compensator be switched ●
- flux valve and directional gyro
 - compass coupler and compass controller
 - flux valve and compass coupler
37. A meter is put on the supply output of a remote compensator. Which kind of voltage is there?
- an extremely constant three-phase voltage
 - a variable low frequency AC voltage
 - an extremely constant DC voltage

38. How is the compensation device carrying the B and C error adjustment screws fitted ●
- it is wired in series with the excitation coil
 - it is screwed into the compass coupler
 - it is screwed onto the flux valve
39. A compensation device is to be fitted to a flux valve. In which case is this done? ●
- only in the case of a failed remote compensator
 - only in the case of persistent interference fields
 - only in the case of a drifting directional gyro
40. A newer type of compass system is fitted. Where is the fuse for the flux valve excitation transformer additionally installed?
- in the flux valve housing
 - in the compass controller
 - in the compass coupler
41. The synchronising annunciator is deflected. Along which path is the difference being reported? ●
- DG – coupler – autoflight system
 - flux valve – coupler – HSI
 - flux valve – coupler – DG
42. Which two signals are coupled together in the compass coupler in the slaved mode ●
- the flux valve signal and the selected VOR course
 - the flux valve signal and the DG platform heading
 - the DG platform heading and the autoflight heading
43. Flux valve and gyro headings disagree in a coupler system. Which device balances them out? ●
- a differential synchro
 - a slaving torque motor
 - a control transformer
44. How does a coupler system compare with the ADF/RMI system for indication behaviour and precision? ●
- it is less precise but reacts faster
 - it is more precise but reacts slower
 - there is no difference between them
45. The system is in its run-up phase. What happens to the update rate of the differential synchro? ●
- it is increased to 700°/min.
 - it is reduced to 0.2°/min.
 - it is held at 2°/min.
46. Which source provides the directional information in the DG mode ●
- only the short-term stable gyro platform
 - only the long-term stable flux valve
 - the gyro platform and the flux valve together
47. What is the state of the synchronising annunciator in the DG mode ●
- it is switched off
 - it shows the drift rate
 - it stands at its centre

48. How does the fast synchronisation control logic raise the update rate ●
- by switching in a second torque motor
 - by bypassing the differential synchro
 - by switching the amplification factor
49. After which length of switch-off pause does switching on in SLAVED mode start the fast synchronisation
- about 2 seconds
 - about 2 minutes
 - about 10 seconds

LEVEL 3 — APPLY AND REASON

50. What is gained by holding the torquing rate of the directional gyro to its printed maximum ●
- drift is still compensated but accelerations cannot move the indication
 - alignment is faster but the gyro cannot be moved by its own drift
 - turns are still compensated but drift cannot move the indication
51. An aircraft is in a long accelerating climb and the flux valve signal is briefly false. Why does the indicated heading stay good? ●
- the flux valve signal is cut off automatically whenever speed changes
 - the torque motor reverses and drives the gyro back to its old heading
 - the gyro can be torqued only slowly, so a brief error cannot move it
52. In the cruise the slaving annunciator keeps fluctuating around zero. What does this tell the pilot?
- the gyro is still running up and the heading is not yet usable at all
 - the gyro has lost its stable direction and the valve is being ignored
 - the gyro has the correct stable direction and the valve is correcting it
53. The slaving annunciator shows a long term deflection to one side. What should be concluded?
- there is an error in the system
 - the gyro drift is being corrected normally
 - the flux valve is seeing an acceleration error
54. Why can the compass rose of the ADF RMI show any heading just after power up ●
- the flux valve needs several minutes before it delivers any signal
 - the slaving annunciator is held at zero during the whole run-up
 - the direction taken up by the gyro after run-up may be any direction
55. The system has just been switched on and the heading shown is wrong. Why does it take time to become correct? ●
- the slaving torque motor may move the gyro by only 2° per minute
 - the compass rose motor is deliberately damped during the run-up
 - the flux valve must first warm up before it delivers its excitation
56. A pilot uses the synchronizing knob shortly after start up. What is the effect? ●
- the slaving annunciator is switched off for the rest of the flight
 - the time until the heading indication is correct becomes much shorter
 - the torquing rate of the gyro is raised to 700° per minute
57. Why is the flux valve signal coupled with the gyro heading instead of being sent straight to the indicator ●
- it is far too weak to drive the compass rose motor of the indicator
 - it can fluctuate wildly for a short time under the influence of acceleration
 - it is a three-phase signal which the compass card cannot accept at all

58. Which unit supplies which kind of stability to the heading signal of a remote compass with ADF/RMI ●
- long-term stability from the flux valve, short-term from the gyro
 - long-term stability from the coupler, short-term from the gyro
 - long-term stability from the gyro, short-term from the flux valve
59. An aircraft accelerates on take off and then cruises for hours. Which unit takes out which error? ●
- the coupler takes out the acceleration errors and the valve the drift
 - the valve takes out the acceleration errors and the gyro the drift
 - the gyro takes out the acceleration errors and the valve the drift
60. An engineer does not want to wait for the automatic alignment. How is the stator of the control transformer adjusted? ●
- by holding the mode switch in the SLAVED position
 - by depressing and rotating the synchronisation knob
 - by depressing and rotating the course selector knob
61. Which errors are offset by the two precision potentiometers of the remote compensator
- the A error alone
 - the B and C errors
 - the A and B errors
62. A flux valve is indexed, that is prefixed. What does this mean when the A error is to be compensated?
- it must not be twisted in its mounting
 - it must be twisted only with the compensator switched off
 - it must be twisted until the B error is also at zero
63. What makes it unnecessary to update the gyro by torquing in a system with a compass coupler ●
- the gyro used in such a system has no drift to be corrected
 - the differences are balanced out by a differential synchro
 - the flux valve signal is passed to the indicator on its own
64. What becomes of the two signals in the coupler when the DG mode is selected ●
- the heading signal still passes through, the valve signal is switched off
 - the heading signal is switched off, the valve signal still passes through
 - the heading signal still passes through, the valve signal is amplified more
65. The flux valve fails during flight. Which mode of operation has to be used? ●
- the DG mode
 - the fast synchronisation mode
 - the slaved mode
66. The system has been in DG mode for a long time and the heading has slowly wandered. What must the crew do? ●
- wait for the synchronising annunciator to null it out
 - switch briefly to the slaved mode and straight back again
 - correct the indication with the SET HDG knob
67. A remote compass system is found in which the gyro is never torqued. Which arrangement must it have? ●
- an indicator in which a control transformer balances the signals
 - a coupler in which a torque motor balances the signals
 - a coupler in which a differential synchro balances the signals

68. Fast synchronisation has been running for well over its minimum time. What else must happen before it is complete?
- the synchronisation error must have come down to 2° per minute
 - the synchronisation error must have come down to 2° or less
 - the synchronising annunciator must have been off for 2 minutes
69. Which state of the fast synchronisation is required for the superflag signal to appear
- fast synchronisation must have been run at least once
 - fast synchronisation must not be in operation
 - fast synchronisation must be in operation at the time
70. Which two conditions concerning the gyro and the servo circuit must hold for the superflag signal to appear
- the DG or INS is valid and the servo circuit is nulled out
 - the DG or INS is valid and the servo circuit is complete
 - the DG is in free azimuth and the servo circuit is complete
71. The maintenance monitor responded only briefly during a flight. What does that mean for the compass system?
- its heading output must be treated as invalid
 - it must be put into the DG mode for the rest of the flight
 - its functioning is not affected
72. A compass system is switched off for half a minute and then switched on again in SLAVED. Does fast synchronisation run? ●
- no, the switch-off pause was too short for it
 - no, it runs only when the mode is left in DG
 - yes, every switch-on in SLAVED will start it
73. After a changeover from DG to SLAVED the heading lines up within seconds instead of minutes. Why is that? ●
- the update rate has been raised to 700°/min. for the changeover
 - the update rate has been raised to 20°/min. for the changeover
 - the gyro has been torqued at its full rate for the changeover

Answers — Remote compass systems and slaving

- a** Those three positions are the ones printed for the flux valve on an aeroplane. (L1, PDF p. 771)
- b** Those are the printed usual installation places for the flux valve of a helicopter. (L1, PDF p. 773)
- c** Drift is the printed disadvantage of the gyro based direction instruments. (L1, PDF p. 775)
- b** Accelerations and turns are what falsify the magnetic sensing devices. (L1, PDF p. 775)
- a** ADF is Automatic Direction Finder and RMI is Radio Magnetic Indicator. (L1, PDF p. 777)
- c** The ADF arrows point at whichever broadcast stations have been tuned. (L1, PDF p. 777)
- c** Those three are the units listed for the remote compass with ADF/RMI. (L1, PDF p. 779)
- b** Weak signals such as the compass rose motor voltage are amplified in the IIS rack. (L1, PDF p. 779)
- b** The screws on the compensation device take out the B and C errors. (L1, PDF p. 781)
- b** The A error is taken out by rotating the valve itself, not by a screw or a potentiometer. (L1, PDF p. 781)
- a** Those four are the units listed for the coupler based system. (L1, PDF p. 785)
- a** The SET HDG knob is the one for setting a heading of choice. (L1, PDF p. 785)
- a** The fast synchronisation logic sits inside the coupler. (L1, PDF p. 787)

14. **b** Fast synchronisation runs at seven hundred degrees a minute. (L1, PDF p. 787)
15. **b** The locations are chosen to keep magnetic interference fields away from the valve. (L2, PDF p. 771)
16. **a** Distance from the helicopter's own magnetic fields decides where the flux valve goes. (L2, PDF p. 773)
17. **a** The swing lets the vertical component of the earth's field reach the valve. (L2, PDF p. 775)
18. **b** The flux valve and the directional gyro together make the remote compass system. (L2, PDF p. 775)
19. **b** Stable heading comes to the indicator from the directional gyro. (L2, PDF p. 775)
20. **c** The difference signal is fed to a torque motor which moves the gyro. (L2, PDF p. 775)
21. **b** Torque on a gyro acts as precession and so moves the gyro's direction. (L2, PDF p. 775)
22. **b** The one indicator carries both the radio navigation picture and the magnetic heading. (L2, PDF p. 777)
23. **b** The annunciator displays how far the indicated heading is from what the flux valve says. (L2, PDF p. 777)
24. **c** In this architecture the detection happens in the indicator itself. (L2, PDF p. 777)
25. **b** The knob acts on what the compass rose shows. (L2, PDF p. 777)
26. **a** That fuse is there for the excitation transformer of the flux valve. (L2, PDF p. 779)
27. **c** The flux valve passes its direction information on as a three-phase signal. (L2, PDF p. 779)
28. **b** The rotor of the slaving CT is driven directly by the compass card. (L2, PDF p. 779)
29. **b** Agreement between the two gives a null at the control transformer. (L2, PDF p. 779)
30. **b** Unaccelerated straight flight is the condition in which the valve is error free. (L2, PDF p. 779)
31. **a** Two degrees either way is the printed precision of the remote compass system. (L2, PDF p. 779)
32. **b** The amplified error signal drives the slaving torque motor on the gyro. (L2, PDF p. 779)
33. **b** Working off a large error at the slow torquing rate can take as long as ninety minutes. (L2, PDF p. 779)
34. **c** Centre position of the moving coil annunciator means the slaving CD is nulled. (L2, PDF p. 779)
35. **c** The moving coil annunciator reads the slaving amplifier output. (L2, PDF p. 779)
36. **c** The remote compensator sits in the line between the flux valve and the coupler. (L2, PDF p. 781)
37. **c** The compensating supply is a very constant direct voltage. (L2, PDF p. 781)
38. **c** The compensation device is mounted directly on the flux valve. (L2, PDF p. 781)
39. **b** It is reserved for interference fields that will not go away. (L2, PDF p. 781)
40. **c** On newer couplers the protecting fuse is also carried in the coupler. (L2, PDF p. 785)
41. **c** It watches the chain from the valve through the coupler to the gyro heading. (L2, PDF p. 785)
42. **b** Slaved mode brings the valve signal and the gyro heading together in the coupler. (L2, PDF p. 785)
43. **a** The differential synchro in the coupler takes out the difference. (L2, PDF p. 785)
44. **c** The two systems behave and perform the same way. (L2, PDF p. 785)
45. **a** Run-up uses the much higher rate so the system lines up quickly. (L2, PDF p. 785)
46. **a** In DG mode the gyro alone supplies the direction. (L2, PDF p. 785)
47. **a** There is nothing for it to compare in DG mode, so it is switched off. (L2, PDF p. 785)
48. **c** A change of amplification factor is what gives the higher rate. (L2, PDF p. 787)
49. **b** A pause of roughly two minutes before a SLAVED start triggers the logic. (L2, PDF p. 787)
50. **a** The rate is fast enough for drift but too slow for the short acceleration errors to get through. (L3, PDF p. 775)
51. **c** The slow torquing rate means a short lived false signal has no time to shift the indication. (L3, PDF p. 775)
52. **c** Fluctuation about zero is the sign of a healthy system doing its normal correcting. (L3, PDF p. 777)
53. **a** A deflection that lasts, rather than one that fluctuates, means a system error. (L3, PDF p. 777)
54. **c** The rose follows the gyro, and the gyro runs up pointing anywhere. (L3, PDF p. 777)
55. **a** A large initial error can only be worked off at the slow torquing rate. (L3, PDF p. 777)
56. **b** Minimising the difference by hand saves the slow automatic torquing. (L3, PDF p. 777)
57. **b** Acceleration makes the raw valve signal jump about, so the gyro steadies it. (L3, PDF p. 779)
58. **a** The valve holds the long term truth, the gyro the short term steadiness. (L3, PDF p. 779)

- 59. c The gyro covers the short acceleration errors, the flux valve the slow drift. (L3, PDF p. 779)
- 60. b The stator is set by hand with the synchronisation knob. (L3, PDF p. 779)
- 61. b The N – S and E – W settings together deal with the B and C errors. (L3, PDF p. 781)
- 62. a An indexed valve is not allowed to be turned, so that way of taking out the A error is barred. (L3, PDF p. 781)
- 63. b Because the coupler itself removes the difference, the gyro need not be torqued. (L3, PDF p. 785)
- 64. a Only the flux valve signal is cut; the heading still runs through the coupler. (L3, PDF p. 785)
- 65. a With no usable valve signal the system has to run on the gyro alone, which is DG mode. (L3, PDF p. 785)
- 66. c In DG mode nothing corrects the drift automatically, so it is set out by hand. (L3, PDF p. 785)
- 67. c Only the coupler arrangement with a differential synchro removes the need to torque the gyro. (L3, PDF p. 785)
- 68. b Both the time and an error within two degrees are needed before it finishes. (L3, PDF p. 787)
- 69. b One of the listed conditions for the superflag is that fast synchronisation is not running. (L3, PDF p. 787)
- 70. b Validity of the gyro or INS and a complete servo circuit are two of the listed conditions. (L3, PDF p. 787)
- 71. c A temporary response is printed as having no effect on how the system works. (L3, PDF p. 787)
- 72. a A pause of about two minutes is needed, so half a minute does not trigger the logic. (L3, PDF p. 787)
- 73. a Switching DG to SLAVED starts the fast synchronisation, which runs at the much higher rate. (L3, PDF p. 787)

Heading indicators and system interfaces

PDF pages 789–795 · 21 questions

LEVEL 1 — RECALL

74. What shows the pilot that the signal from his compass system is not valid ●
- a. a heading warning flag
 - b. the synchronising annunciator
 - c. the compass superflag
75. What does the abbreviation HSI stand for ●
- a. Horizontal Situation Indicator
 - b. Horizontal Stabiliser Indicator
 - c. Heading and Situation Instrument
76. A pilot glances at the HSI for his heading. Which indication on it is the main one? ●
- a. the distance readout
 - b. the course bar
 - c. the compass rose

LEVEL 2 — UNDERSTAND AND EXPLAIN

77. A pilot reads the heading on the ND in front of him. Which compass system supplies it? ●
- a. their own compass system
 - b. whichever system the transfer switch selects
 - c. the opposite compass system
78. Which compass system do the RMIs show ●
- a. the system feeding the autoflight computer
 - b. the opposite compass system
 - c. their own compass system
79. To which autoflight system is the heading of compass system 1 provided ●
- a. to both autoflight systems together
 - b. to autoflight system number 2
 - c. to autoflight system number 1
80. To what besides the other heading sensor can the heading transfer switch connect a user ●
- a. to a third unit if one is available
 - b. to the flux valve signal taken on its own
 - c. to the autoflight system of the other side
81. A heading leaves the compass coupler on its way to the HSIs. Which heading is it?
- a. the free azimuth platform heading
 - b. the uncorrected flux valve heading
 - c. the gyro supported magnetic heading
82. To which compass system is the flight recorder normally switched ●
- a. to both systems in turn
 - b. to compass system 1
 - c. to compass system 2

- 83.** How are the ADF RMIs normally switched to the two compass systems? ●
- crosswise
 - both to system 1
 - both to system 2
- 84.** From which viewpoint does the HSI present the situation of the aircraft ●
- as seen from the side
 - as seen from above
 - as seen from behind
- 85.** To which kind of ground station is the course selected on an HSI related ●
- an ADF broadcast station
 - a DME ground station
 - a VOR ground radio station
- 86.** The white bar on an HSI has moved away from the centre. What is it telling the crew? ●
- there is a deviation in the distance to the station
 - there is a deviation from the selected heading
 - there is a deviation from the selected course
- 87.** What can the HSI show with respect to a DME ground station ●
- the distance to it
 - the deviation from it
 - the bearing to it
- 88.** Which kind of system do all the further indications on an HSI belong to ●
- navigation systems
 - attitude reference systems
 - autoflight systems

LEVEL 3 — APPLY AND REASON

- 89.** A pilot compares the heading on the ND with the heading on the RMI in front of him. Which sources is he comparing? ●
- his own compass system against the opposite one
 - the opposite compass system against the flight recorder
 - his own compass system against the autoflight system
- 90.** A heading warning flag has appeared on one side, so what can the pilot do about it? ●
- use the synchronizing knob to re-set the compass rose by hand
 - use the heading transfer switch to take the other heading sensor
 - use the compass controller to take the free azimuth heading instead
- 91.** An HSI loses its heading while the RMI beside it still shows one. Which compass system has gone? ●
- the one belonging to the opposite side of the cockpit
 - the one selected with the heading transfer switch
 - the one belonging to that side of the cockpit
- 92.** One of the two compass systems fails in flight. What happens to the consumers that were fed by it?
- they can all be switched to the system that is still working
 - they hold the last valid heading until the system is reset
 - they are fed directly from the flux valve of the failed system

- 93.** What is achieved by exciting the synchros from various voltage sources
- a receiver is always supplied even when the coupler itself has failed
 - a receiver is always fed from two independent buses at the same time
 - a receiver is always on a transmitting synchro fed by its own bus
- 94.** An HSI is showing a compass rose, a course bar and a distance. Which of these comes from the compass system? ●
- the distance
 - the course bar
 - the compass rose

Answers — Heading indicators and system interfaces

- 74. a** Loss of validity is shown by the heading warning flag. (L1, PDF p. 789)
- 75. a** HSI is the Horizontal Situation Indicator. (L1, PDF p. 795)
- 76. c** The compass rose is the HSI's main indication. (L1, PDF p. 795)
- 77. a** The HSI or ND is fed by the compass system of its own side. (L2, PDF p. 789)
- 78. b** The RMI deliberately carries the heading of the other side's system. (L2, PDF p. 789)
- 79. c** The autoflight systems are fed straight across, one for one. (L2, PDF p. 789)
- 80. a** Where a third unit exists the transfer switch can reach it too. (L2, PDF p. 789)
- 81. c** What leaves the coupler is the magnetic heading after the gyro has steadied it. (L2, PDF p. 791)
- 82. c** The flight recorder normally sits on compass system 2. (L2, PDF p. 791)
- 83. a** The ADF RMIs are fed across, each from the other side's system. (L2, PDF p. 791)
- 84. b** The HSI gives a plan view, the aircraft seen from above. (L2, PDF p. 795)
- 85. c** The selectable course on the HSI belongs to a VOR station. (L2, PDF p. 795)
- 86. c** The bar is the course deviation indication of the HSI. (L2, PDF p. 795)
- 87. a** From a DME station the HSI takes the distance. (L2, PDF p. 795)
- 88. a** Every extra indication on the HSI is a navigation system indication. (L2, PDF p. 795)
- 89. a** Because the two displays are fed from opposite systems, the comparison is a cross check of both. (L3, PDF p. 789)
- 90. b** The transfer switch is the printed way of getting a valid heading back. (L3, PDF p. 789)
- 91. c** The HSI is fed by its own system and the RMI by the other one, so the failure is on its own side. (L3, PDF p. 789)
- 92. a** Every affected consumer can be moved over to the surviving system. (L3, PDF p. 791)
- 93. c** Spreading the excitation keeps each receiver and its synchro on a common bus. (L3, PDF p. 791)
- 94. c** The rose is the heading indication; the course bar and the distance are navigation system indications. (L3, PDF p. 795)

Glass cockpit displays

PDF pages 797–807 • 41 questions

LEVEL 1 — RECALL

95. What is meant by a glass cockpit ●
- the instruments are fitted behind one large glass panel
 - the instrument glasses are replaced by a scratch free type
 - computers and displays are used instead of instruments
96. What is the system used for flight monitoring in a glass cockpit called in many aircraft ●
- CDS
 - EFIS
 - EIS
97. Which three items are the components of a typical electronic instrument system ●
- Display units, control panels and pressure sensors
 - Display computers, display units and control panels
 - Display computers, control panels and warning horns
98. A system is fitted with as few display computers as it is allowed to have, so how many is that? ●
- Two
 - Three
 - Four
99. How many indication display screens represent the display system in the cockpit ●
- Four
 - Eight
 - Six
100. What do the letters ECAM stand for ●
- Electronic central alert monitor
 - Electronic centralized aircraft monitor
 - Engine centralized aircraft monitor
101. How many display management computers are installed and what do they do with the data they receive ●
- Two computers, forwarding the data to the display units
 - Three computers, forwarding the data to the warning computers
 - Three computers, forwarding the data to the display units
102. An engineer counts the flight warning computers of the system, so how many does he find? ●
- One
 - Three
 - Two

LEVEL 2 — UNDERSTAND AND EXPLAIN

103. A crew is flying an aircraft with a glass cockpit. How is their flight monitoring done? ●
- by one display for each pilot
 - by two displays shared by both pilots
 - by two displays for each pilot

- 104.** Where are the many engine parameters shown in a glass cockpit ●
- on the related engine control panel
 - on two central displays
 - on two displays for each pilot
- 105.** Where are the alerts of the central warning panel presented in a glass cockpit ●
- on a printed caption board
 - at the related control panel
 - in a display
- 106.** The engine data and the auto flight system data reach a display computer, so in which form do they arrive?
- One is digital and one is analogue
 - Both are digital
 - Both are analogue
- 107.** Why does a large aircraft carry a third display computer ●
- To convert analogue inputs
 - To drive the control panels
 - For redundancy
- 108.** How does the electronic flight instrument system stand in relation to the electronic instrument system ●
- It replaces it and shows alert messages
 - It is a separate system and shows engine data
 - It is a part of it and shows navigation data
- 109.** How many control panels serve the centre displays and what is selected on them ●
- Two panels, used to select the system to be monitored
 - One panel, used to select the computer to be monitored
 - One panel, used to select the system to be monitored
- 110.** Which further information is presented on the electronic instruments besides the navigation data ●
- The condition of the air conditioning system
 - The limits of the engine parameters
 - The activities of the auto flight system
- 111.** How are the six indication display screens divided between the electronic flight instruments and the central monitoring displays ●
- Two flight instrument displays and four monitoring displays
 - Three flight instrument displays and three monitoring displays
 - Four flight instrument displays and two monitoring displays
- 112.** Which two displays make up the centralized monitoring pair on an Airbus aircraft ●
- The engine and warning display and the system display
 - The engine and warning display and the navigation display
 - The primary flight display and the system display
- 113.** Which of the two EICAS displays on a Boeing aircraft carries the engine and warning indications ●
- The auxiliary EICAS display
 - The upper EICAS display
 - The lower EICAS display

- 114.** Which kind of information does the primary flight display supply ●
- Flight monitoring information and parameters
 - Flight planning information and parameters
 - Engine monitoring information and parameters
- 115.** Which parameter is listed among those supplied by the primary flight display? ●
- Fuel on board
 - Hydraulic pressure
 - Radio altitude
- 116.** Which annunciation is listed among the parameters of the primary flight display ●
- The flight mode annunciation
 - The engine mode annunciation
 - The flight phase annunciation
- 117.** With respect to which plane does the navigation display present the situation of the aircraft ●
- The horizontal plane with respect to the flight path
 - The horizontal plane with respect to the earth's surface
 - The vertical plane with respect to the earth's surface
- 118.** A flight warning computer works on the data it receives, so which three kinds of output does it generate? ●
- Alert messages, synthetic voice and engine parameters
 - Alert messages, aural alerts and system pages
 - Alert messages, aural alerts and synthetic voice
- 119.** An announcement in spoken words is heard in the cockpit, so which of the three warning outputs is this? ●
- Aural alert
 - Synthetic voice
 - Alert message
- 120.** Which computer decides what is to appear on the centralized monitoring displays ●
- The system data acquisition concentrator
 - The display management computer
 - The flight warning computer
- 121.** To which two sets of computers does a system data acquisition concentrator forward its data ●
- The flight warning computers and the display management computers
 - The display management computers and the control panels
 - The flight warning computers and the other concentrator
- 122.** Which computer puts the engine parameters and the system page parameters onto the displays? ●
- The system data acquisition concentrator
 - The display management computer
 - The flight warning computer
- 123.** Which three kinds of warning are generated by the modularized avionics warning electronics assembly ●
- Optical, aural and tactile warnings
 - Optical, tactile and synoptic warnings
 - Optical, aural and visual warnings

124. Through which item of equipment are the warnings of the avionics warning electronics assembly generated ●
- Through the control panel
 - Through the display select panel
 - Through the interface unit
125. Which three items are controlled by the avionics warning electronics assembly itself? ●
- The master warning light, the aural warn speaker and the display units
 - The master warning light, the aural warn speaker and the stick shaker
 - The master caution light, the aural warn speaker and the stick shaker

LEVEL 3 — APPLY AND REASON

126. An aircraft without a glass cockpit is being compared with one that has it. What does flight monitoring use on the older aircraft? ●
- two displays for each pilot and some other navigation indicators
 - two central displays and the indicators of a central warning panel
 - the basic T instruments and some other navigation indicators
127. On which display is the crew told that a value somewhere in the aircraft has gone out of limit
- On one of the two engine and system data displays
 - On the primary flight display of each pilot
 - On the navigation display of each pilot
128. An aircraft is fitted with only the minimum number of display computers, so what does it lack compared with a large aircraft? ●
- The computer that handles the analogue inputs
 - The computer that drives the centre displays
 - The computer that is there for redundancy
129. Which part inside a second generation display unit makes a picture out of the data it is sent
- Its integrated display computer
 - Its integrated symbol generator
 - Its integrated control panel
130. Which Boeing display does the same job as the Airbus system display ●
- The auxiliary EICAS display
 - The upper EICAS display
 - The engine and warning display
131. Which path is taken by the signal that produces an amber caution ●
- From the sensors and computers direct to the warning computer
 - From the sensors and computers through the data concentrator to the warning computer
 - From the sensors and computers through the display computer to the warning computer
132. Both system data acquisition concentrators are lost, so which colour of warning can still be generated? ●
- The amber cautions
 - The red warnings
 - Neither colour

- 133.** The engine and warning screen stays blank but the aural alerts are still heard, so which computer is suspected? ●
- The display management computer
 - The flight warning computer
 - The data acquisition concentrator
- 134.** The stick shaker does not operate although the conditions for it are met, so which unit is first suspected? ●
- The avionics warning electronics assembly
 - The EICAS display select panel
 - The EFIS and EICAS interface unit
- 135.** The master warning light comes on but no message reaches the screens, so which item is suspected? ●
- The EICAS display select panel
 - The avionics warning electronics assembly
 - The EFIS and EICAS interface unit

Answers — Glass cockpit displays

- 95. c** Glass cockpit means instruments have given way to computers and displays. (L1, PDF p. 797)
- 96. b** The flight monitoring displays are commonly called EFIS. (L1, PDF p. 797)
- 97. b** The three listed components are the display computers, the display units and the control panels. (L1, PDF p. 799)
- 98. a** The text gives at least two display computers as the minimum. (L1, PDF p. 799)
- 99. c** The cockpit carries six indication display screens. (L1, PDF p. 801)
- 100. b** ECAM is the electronic centralized aircraft monitor. (L1, PDF p. 801)
- 101. c** Three identical display management computers take in the data and pass it to the display units. (L1, PDF p. 805)
- 102. c** Two identical flight warning computers are fitted. (L1, PDF p. 805)
- 103. c** Each pilot has two displays for flight monitoring. (L2, PDF p. 797)
- 104. b** Engine parameters go on the two central displays. (L2, PDF p. 797)
- 105. c** Those alerts move from the panel onto a display. (L2, PDF p. 797)
- 106. b** Navigation, auto flight and engine inputs are all listed as digital. (L2, PDF p. 799)
- 107. c** The third computer is fitted on large aircraft for redundancy. (L2, PDF p. 799)
- 108. c** EFIS is part of the EIS and its job is to show navigation data. (L2, PDF p. 799)
- 109. c** A single control panel serves the centre displays and selects the system to be monitored. (L2, PDF p. 799)
- 110. c** The electronic flight instruments also show what the auto flight system is doing. (L2, PDF p. 799)
- 111. c** Four of the six are flight instrument displays and two are central monitoring displays. (L2, PDF p. 801)
- 112. a** The pair is the engine and warning display together with the system display. (L2, PDF p. 801)
- 113. b** The upper EICAS is the engine and warning display and the auxiliary one is the system display. (L2, PDF p. 801)
- 114. a** The primary flight display is the flight monitoring display. (L2, PDF p. 801)
- 115. c** Radio altitude is one of the parameters listed for the primary flight display. (L2, PDF p. 801)
- 116. a** The flight mode annunciation is one of the listed primary flight display parameters. (L2, PDF p. 801)
- 117. b** The navigation display works in the horizontal plane referred to the earth's surface. (L2, PDF p. 801)
- 118. c** The three outputs are the optical alert messages, the acoustic aural alerts and the voice announcements. (L2, PDF p. 805)
- 119. b** The voice announcement is the synthetic voice output, while the acoustic one is the aural alert. (L2, PDF p. 805)
- 120. c** The flight warning computer determines the content of the ECAM displays. (L2, PDF p. 805)
- 121. a** The concentrator feeds both the flight warning computers and the display management computers. (L2, PDF p. 805)
- 122. b** Overlaying the engine and system page parameters is the display management computer's job. (L2, PDF p. 805)

- 123. a** It produces optical, aural and tactile warnings. (L2, PDF p. 807)
- 124. c** The warnings are generated by way of the EFIS/EICAS interface unit. (L2, PDF p. 807)
- 125. b** Those three are the items it drives directly. (L2, PDF p. 807)
- 126. c** Without displays the job is done by the basic T instruments plus other navigation indicators. (L3, PDF p. 797)
- 127. a** One of the two engine and system displays additionally carries the alerts for out of limit conditions. (L3, PDF p. 799)
- 128. c** The minimum fit is two computers, so the third one carried for redundancy on large aircraft is missing. (L3, PDF p. 799)
- 129. b** These units are sent data rather than video, so the symbol generator built into them draws the picture. (L3, PDF p. 799)
- 130. a** The auxiliary EICAS does the job that the system display does on the other manufacturer's fit. (L3, PDF p. 801)
- 131. b** An amber caution reaches the flight warning computer by way of the system data acquisition concentrator. (L3, PDF p. 805)
- 132. b** Red warnings reach the warning computer without passing through the concentrator, so they survive its loss. (L3, PDF p. 805)
- 133. a** The warnings are still being generated, so the fault lies in the computer that feeds the display unit. (L3, PDF p. 805)
- 134. a** The stick shaker is one of the items the warning electronics assembly controls directly. (L3, PDF p. 807)
- 135. c** The light is driven directly and works, so the path through the interface unit to the screens is at fault. (L3, PDF p. 807)

Display management and failures

PDF pages 811–823 · 24 questions

LEVEL 1 — RECALL

- 136.** What appears on the display in place of a parameter whose input signal is not available ●
- a. A flag
 - b. An amber box
 - c. A dash

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 137.** What does a cathode ray tube display unit do first when it detects an internal overheat ●
- a. It switches the display off
 - b. It switches the strokes off
 - c. It switches the raster colours off
- 138.** What is left on the screen after the first step of the overheat reaction ●
- a. Only the raster, with all information still given
 - b. Only strokes, with the warnings removed
 - c. Only strokes, with all information still given
- 139.** Why does switching the raster colours off help a display unit that is running hot? ●
- a. Less light is needed from the backlight
 - b. Less heat is produced then
 - c. Less current is drawn by the cooling fan
- 140.** What does the display unit do as the second step of its overheat reaction ●
- a. It switches the display to the cross side computer
 - b. It switches the raster colours off
 - c. It switches the display off
- 141.** Which two causes are given for an overheat of a display unit? ●
- a. An internal failure or a problem in the power supply
 - b. A high cabin temperature or a problem in the cooling system
 - c. An internal failure or a problem in the cooling system
- 142.** What happens to the white in the picture when one electron beam of a cathode ray tube fails ●
- a. There is no white in the picture
 - b. The white is replaced by amber in the picture
 - c. The white turns to grey in the picture
- 143.** What becomes of the information that is normally presented in the colour of a failed electron beam ●
- a. It is shown as a flag instead
 - b. It is shown in white instead
 - c. It will not be shown
- 144.** What does the display do when the red electron beam fails ●
- a. It loses only the white
 - b. It switches itself off
 - c. It turns monochrome

- 145.** Which electron beam has failed when the crew sees the whole display in one colour ●
- The blue beam
 - The red beam
 - The green beam
- 146.** Which part of the display is affected when one input signal to a display unit is lost ●
- Only the indication of the related parameter
 - The whole of the display unit
 - Only the warning labels of the display

LEVEL 3 — APPLY AND REASON

- 147.** Where does a helicopter crew find the engine alerts and where the navigation picture
- Both of them on the centre instrument panel
 - The alert on the outboard panels and the picture on the centre panel
 - The alert on the centre panel and the picture on the outboard panels
- 148.** Where does the information of a primary flight display appear when that display goes blank in flight
- On the navigation display
 - On the opposite primary flight display
 - On the centre engine display
- 149.** A pilot's primary flight display goes dark, so what must he do before the information comes back?
- Select the cross side computer on his control panel
 - Nothing, because the failed screen restarts on its own
 - Nothing, because the transfer happens on its own
- 150.** A display unit that switched itself off because of overheat has cooled down again, so what happens next? ●
- The picture comes back without any crew action
 - The picture stays off until the crew resets it
 - The picture comes back without the raster colours
- 151.** Which failure fits a screen that has lost its colours but still shows every item of information ●
- A total loss of the display unit
 - The first step of the overheat reaction
 - The second step of the overheat reaction
- 152.** What comes next when a screen has already dropped its raster colours and the cause has not been cleared ●
- The screen drops its strokes as well
 - The screen switches to the cross side computer
 - The screen switches itself off
- 153.** A pilot reports that the green symbols have gone and nothing on the screen looks white any more, so what has happened? ●
- One electron beam of the tube has failed
 - The raster colours have been switched off
 - An input signal to the display has been lost

- 154.** Why is the display turned monochrome rather than left with its remaining colours when the red beam fails? ●
- Because the warning labels are red and must not be allowed to disappear
 - Because the remaining colours would produce too much heat
 - Because the warning labels are red and must not be allowed to flicker
- 155.** Why is a failure of the red beam treated differently from the failure of another beam? ●
- Because the information it carries must never be lost
 - Because it carries more of the picture than the others
 - Because it produces more heat than the others
- 156.** Two screens have lost a beam and only one of them went monochrome, so what separates the two cases? ●
- The monochrome one lost the beam that draws the green symbols
 - The monochrome one lost the beam that draws the warnings
 - The monochrome one lost two beams instead of one
- 157.** What is wrong when a single scale on the primary flight display is replaced by a flag? ●
- The input signal for that parameter is not available
 - The display unit has started its overheat reaction
 - One electron beam of the display tube has failed
- 158.** Several indications are gone and flags stand in their place, so what does this tell the engineer? ●
- The display unit has an electron beam failure
 - The signals for those parameters are missing
 - The display unit has begun to overheat
- 159.** One screen has lost some of its symbols and another shows flags in their place, so are the causes the same? ●
- Yes, both of them come from a failure inside the display unit
 - No, the flags come from missing signals and the lost symbols from a failed beam
 - No, the flags come from a failed beam and the lost symbols from missing signals

Answers — Display management and failures

- 136. a** The indication disappears and a flag is shown instead. (L1, PDF p. 823)
- 137. c** The first step is to switch off the raster colours. (L2, PDF p. 817)
- 138. c** Only the strokes remain but none of the information is lost. (L2, PDF p. 817)
- 139. b** Dropping the raster colours makes the tube produce less heat. (L2, PDF p. 817)
- 140. c** The second step is to switch the display off altogether. (L2, PDF p. 817)
- 141. c** The two causes given are an internal failure and problems in the cooling system. (L2, PDF p. 817)
- 142. a** With one beam gone the picture contains no white at all. (L2, PDF p. 819)
- 143. c** Information carried by the missing colour simply disappears from the screen. (L2, PDF p. 819)
- 144. c** A failed red beam makes the whole display go monochrome. (L2, PDF p. 821)
- 145. b** Only the failure of the red beam makes the display go monochrome. (L2, PDF p. 821)
- 146. a** Only the indication belonging to the missing signal goes away. (L2, PDF p. 823)
- 147. c** Engine and warning indications sit in the centre while the flight and navigation displays sit in front of the crew. (L3, PDF p. 811)
- 148. a** The information of a failed primary flight display is switched automatically onto the navigation display. (L3, PDF p. 813)
- 149. c** The takeover by another screen is automatic, so no crew action is needed to get the information back. (L3, PDF p. 813)

- 150. a** Once it is back within the limits the unit activates its display again on its own. (L3, PDF p. 817)
- 151. b** Losing the raster colours while keeping all the information is the first overheat step. (L3, PDF p. 817)
- 152. c** The two step reaction goes on to its second step, in which the display is switched off. (L3, PDF p. 817)
- 153. a** Loss of one beam removes both the white and everything drawn in that beam's colour. (L3, PDF p. 819)
- 154. a** Warnings are drawn in red, so losing that beam would hide them unless the whole picture is made monochrome. (L3, PDF p. 821)
- 155. a** The red beam draws the warning labels, which may not be allowed to disappear. (L3, PDF p. 821)
- 156. b** Only the loss of the beam that draws the red warning labels turns the whole picture monochrome. (L3, PDF p. 821)
- 157. a** A flag in place of one parameter is the sign that the input signal for it is missing. (L3, PDF p. 823)
- 158. b** Flags are the signature of missing input signals, not of a fault inside the display unit. (L3, PDF p. 823)
- 159. b** A flag marks a missing input, while symbols that vanish without a flag point to a beam failure in the tube. (L3, PDF p. 823)

Head up display and EFIS layout

PDF pages 825–839 · 18 questions

LEVEL 1 — RECALL

- 160.** Which display units normally show the ND contents ●
- a. The lower display unit
 - b. The inboard display units
 - c. The outboard display units

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 161.** Which movement of the pilot is avoided by the use of a head up display
- a. Lowering his head from the cockpit window to the control pedestal
 - b. Lowering his head from the cockpit window to the displays below
 - c. Turning his head from the cockpit window to the overhead panel
- 162.** Under which condition is the head up display meant to give guidance information on the ground and at take-off and landing ●
- a. When the landing gear is extended
 - b. When the autopilot is engaged
 - c. When visibility is reduced
- 163.** Which automatic procedure is the head up display used to monitor? ●
- a. The auto approach
 - b. The auto land flare
 - c. The auto thrust reduction

LEVEL 3 — APPLY AND REASON

- 164.** How does the head up display help the pilot at an airfield that has no instrument landing system ●
- a. It gives the pilot the glide slope deviation
 - b. It gives the pilot the localizer deviation
 - c. It gives the pilot information for the visual approach
- 165.** Visibility is poor on the approach, so why can the pilot keep his eyes on the outside world? ●
- a. The information is presented on a screen at head height
 - b. The information is read out to him by a synthetic voice
 - c. The information is presented on a screen below the window
- 166.** Why does the picture seen on the head up display change as the aircraft goes from take-off to cruise?
- a. Because the display depends on the selected control panel
 - b. Because the display depends on the flight configuration
 - c. Because the display depends on the available input signals
- 167.** Which segment of the head up display image is used while the automatic system flies the aircraft down to the runway
- a. The ground roll information segment
 - b. The approach trajectory deviation segment
 - c. The system messages segment

- 168.** Why a parameter can be absent from a display at one point in the flight and present at another
- Because much data is only shown in the flight phase it belongs to
 - Because the control panel is reset between the flight phases
 - Because the display units are tested one after the other in flight
- 169.** A mechanic finds a glass cockpit aircraft with no separate DME or marker indicator fitted, so what does that tell him?
- Those indications are only on the standby panel
 - Those indications are handled by the EFIS
 - Those two systems are not installed
- 170.** A trainee asks why a glass cockpit needs no separate flight mode annunciator, so what is the answer?
- The ND carries that zone itself
 - The PFD carries that zone itself
 - The control panel carries that zone itself
- 171.** Which display is affected by four of the five basic EFIS control panel functions
- The lower display unit
 - The PFD
 - The ND
- 172.** A pilot selects TCAS on the EFIS control panel, so where does the traffic picture appear?
- On the standby instrument
 - On the PFD
 - On the ND
- 173.** A mechanic moves to a new aircraft type and finds an unfamiliar switch on its EFIS control panel, so how should he treat it?
- As a basic function common to every type
 - As a type specific additional function
 - As a function that acts on the standby instruments
- 174.** An outboard display unit goes blank in flight, so what happens to the PFD picture? ●
- It is moved to the standby attitude instrument
 - It is lost until the display unit is replaced
 - It can be switched to the inboard display unit
- 175.** In which two ways the PFD can reach the inboard display unit after an outboard display failure ●
- Manually from either seat
 - Automatically or manually
 - Automatically after a delay
- 176.** A pilot wants his navigation picture on the lower display unit, so how does he get it there? ●
- It appears there automatically
 - He selects it on the source select module
 - He switches it there manually
- 177.** Both outboard display units fail, so what does the crew lose on the inboard screens? ●
- The normal navigation pictures
 - The normal warning indications
 - The normal engine indications

Answers — Head up display and EFIS layout

- 160. b** The two inboard display units normally carry the navigation display contents. (L1, PDF p. 839)
- 161. b** The pilot no longer has to lower his head to the conventional displays below the window. (L2, PDF p. 825)
- 162. c** That guidance is provided for the case of reduced visibility. (L2, PDF p. 825)
- 163. a** One stated purpose is to monitor the auto approach. (L2, PDF p. 825)
- 164. c** Where no instrument landing system exists the head up display supports the visual approach. (L3, PDF p. 825)
- 165. a** Because the screen sits at head height he can read the guidance without looking down into the cockpit. (L3, PDF p. 825)
- 166. b** The pilot is given different displays according to the flight configuration. (L3, PDF p. 827)
- 167. b** Monitoring an automatic approach calls for the segment that shows how far the aircraft is off the trajectory. (L3, PDF p. 827)
- 168. a** The flexibility of the screen lets much data be displayed only during the respective flight phase. (L3, PDF p. 831)
- 169. b** The DME indicator and the marker lights are both in the list of instruments the EFIS replaces. (L3, PDF p. 831)
- 170. b** The flight mode annunciator is replaced by the EFIS and appears as a zone of the PFD. (L3, PDF p. 831)
- 171. c** Mode, range, TCAS and additional information selections all act on the ND, leaving only baro correction for the PFD. (L3, PDF p. 833)
- 172. c** Selection of the TCAS indication is a basic function and the indication appears on the ND. (L3, PDF p. 833)
- 173. b** Beyond the common basic functions, many types carry some type specific additional functions. (L3, PDF p. 833)
- 174. c** On failure of the outboard display the PFD can be switched automatically or manually to the inboard DU. (L3, PDF p. 839)
- 175. b** The switching to the inboard DU can be done automatically or manually. (L3, PDF p. 839)
- 176. c** The captain's or F/O's ND can be switched onto the lower display unit manually. (L3, PDF p. 839)
- 177. a** The inboard units normally show the ND, so taking the PFDs onto them costs the crew those navigation pictures. (L3, PDF p. 839)

EFIS displays and symbols

PDF pages 841–852 · 25 questions

LEVEL 1 — RECALL

- 178.** On which display unit is the PFD normally shown ●
- The inboard pair
 - The lower one
 - The outboard one
- 179.** From which viewpoint the navigation display presents the aircraft situation ●
- From above
 - From behind
 - From the side
- 180.** Which conventional instrument is fully replaced by the navigation display ●
- The HSI
 - The ADI
 - The RMI
- 181.** During which part of the flight the ND ILS mode indicates its parameters ●
- During the taxi out
 - During the ILS approach
 - During the initial climb out
- 182.** A pilot selects the ILS mode on his ND, so which two deviations does he then see? ●
- Glideslope and vertical speed
 - Localizer and track
 - Localizer and glideslope
- 183.** Which three items are added in the navigation sector mode of the ND ●
- The range selection, the localizer scale and the waypoint
 - The waypoint, the glideslope scale and the route indication
 - The waypoint, the range selection and the route indication

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 184.** A pilot's PFD appears on the inboard screen in flight, so what has most likely happened? ●
- The baro correction has been changed
 - The control panel has been reset
 - The normal display unit has failed
- 185.** What the small aircraft symbol on the navigation display is there for ●
- To show the aircraft type being flown
 - To act as a reference for the picture
 - To show the attitude of the aircraft
- 186.** Which navigation point does the basic waypoint data on the ND refer to ●
- The last one passed
 - The destination one
 - The next one

- 187.** A trainee asks what ground speed on the navigation display means, so what is the answer? ●
- The speed limit on the ground
 - The speed through the air
 - The speed above the ground
- 188.** Which ND parameter comes from somewhere other than the navigation systems ●
- The heading
 - The ground speed
 - The wind velocity
- 189.** Which aircraft system supplies the parameters shown in the ND ILS mode ●
- The flight management computer system
 - The air data computer system
 - The instrument landing system
- 190.** A route indication is missing from a navigation display in the sector mode, so which system should be suspected? ●
- The flight management system
 - The ground proximity warning system
 - The instrument landing system

LEVEL 3 — APPLY AND REASON

- 191.** Why the loss of one display unit need not cost the pilot his primary flight picture ●
- Because the PFD can be switched to another display unit
 - Because the PFD is drawn on a second display unit as well
 - Because the standby instruments take over the PFD picture
- 192.** Which system supplies the AFCS information shown on the PFD
- The navigation system
 - The autoflight system
 - The air data system
- 193.** The pressure setting on a PFD will not change, so which control should be suspected first?
- The range selection on the EFIS control panel
 - The baro correction on the EFIS control panel
 - The mode selection on the EFIS control panel
- 194.** Which two deviations are shown together on the PFD in the ILS mode
- The localizer and the vertical speed
 - The glide slope and the radio height
 - The localizer and the glide slope
- 195.** A mechanic is asked why there is no separate ILS indicator in an EFIS cockpit, so what should he say?
- The ILS picture is only available on the ground
 - The PFD takes that job in the ILS mode
 - The ILS is only monitored from the standby instrument
- 196.** Why the wind indication stays on the navigation display whatever mode is selected ●
- Because the control panel cannot reach that part of the screen
 - Because it is only drawn by the air data computer
 - Because it is one of the basic items always present

- 197.** A pilot changes the ND mode and loses an indication he wanted, so what does that tell him about it? ●
- It is not one of the basic always present items
 - It is only driven by the flight management system
 - It is only shown while the autopilot is engaged
- 198.** The navigation systems feeding an ND are lost but the heading is still drawn, so what does that say about the heading? ●
- It is drawn by the flight management computer
 - It is navigation data like every one of the others
 - It is the one item that is not navigation data
- 199.** Why the localizer deviation can be read on the PFD and on the ND at the same time ●
- Because both displays can present ILS parameters
 - Because the ND copies the whole PFD picture in that mode
 - Because the ILS feeds only the navigation display units
- 200.** Why selecting the navigation sector mode changes what the pilot can read on the display ●
- Because that mode adds flight management data
 - Because that mode removes the compass rose from the picture
 - Because that mode moves the picture onto the lower display unit
- 201.** A pilot swaps from the ILS mode to the navigation sector mode, so what replaces the ILS parameters on his display? ●
- Flight management information
 - Radio altimeter information
 - Inertial reference information
- 202.** A pilot reduces the display range on the ND, so what happens to the small aircraft symbol?
- It moves towards the edge of the screen
 - It is removed until the range is restored
 - It stays in the centre of the screen

Answers — EFIS displays and symbols

- 178. c** The PFD is normally presented on the outboard display unit. (L1, PDF p. 841)
- 179. a** The ND shows the situation of the aircraft seen from above. (L1, PDF p. 847)
- 180. a** The ND fully replaces the horizontal situation indicator. (L1, PDF p. 847)
- 181. b** The ILS mode indicates ILS parameters during the ILS approach. (L1, PDF p. 849)
- 182. c** The ILS mode of the ND indicates localizer and glideslope deviation. (L1, PDF p. 849)
- 183. c** The nav sector mode additionally displays waypoint, range selection and route indication. (L1, PDF p. 851)
- 184. c** The PFD is switched to the inboard display unit on failure of its normal one. (L2, PDF p. 841)
- 185. b** The small aircraft symbol is used as a reference and is normally in the centre of the screen. (L2, PDF p. 847)
- 186. c** The waypoint data gives information about the next navigation point. (L2, PDF p. 847)
- 187. c** Ground speed is printed as the speed above the ground. (L2, PDF p. 847)
- 188. a** Apart from the heading, all information on the ND is based on data from the navigation systems. (L2, PDF p. 847)
- 189. c** The ILS mode indicates parameters coming from the instrument landing system. (L2, PDF p. 849)
- 190. a** The route indication in that mode is supplied by the flight management system. (L2, PDF p. 851)
- 191. a** The PFD can be switched automatically or manually to the inboard display unit when its own unit fails. (L3, PDF p. 841)
- 192. b** The AFCS information on the PFD comes from the autoflight system. (L3, PDF p. 843)

- 193. b** The pressure setting on the PFD is the baro correction, which is set from the EFIS control panel. (L3, PDF p. 843)
- 194. c** The ILS mode displays localizer and glide slope alongside the source and the marker. (L3, PDF p. 845)
- 195. b** The PFD itself is used as the display for the instrument landing system. (L3, PDF p. 845)
- 196. c** Wind direction and velocity is one of the basic pieces of information present no matter what is selected. (L3, PDF p. 847)
- 197. a** Only the listed basic pieces of information are present no matter what is selected on the control panel. (L3, PDF p. 847)
- 198. c** Apart from the heading, all information on the ND is based on data from the navigation systems. (L3, PDF p. 847)
- 199. a** The PFD has an ILS mode and the ND has an ILS mode, and both show the localizer indication. (L3, PDF p. 849)
- 200. a** The nav sector mode adds the waypoint, range selection and route indication, all from the FMS. (L3, PDF p. 851)
- 201. a** The ILS mode shows ILS parameters while the nav sector mode adds information from the flight management system. (L3, PDF p. 851)
- 202. c** The small aircraft symbol is used as a reference and is normally displayed in the centre of the screen. (L3, PDF p. 852)

Central warning systems

PDF pages 859–871 • 67 questions

LEVEL 1 — RECALL

- 203.** How do the pushbuttons appear while everything is operating normally ●
- a. Steady white
 - b. Steady green
 - c. Dark
- 204.** Which colour is used when an immediate action is necessary ●
- a. Amber
 - b. Red
 - c. White
- 205.** Which colour indicates the normal operation of a backup system ●
- a. White
 - b. Green
 - c. Blue
- 206.** Which colour indicates the normal operation of a temporarily used system ●
- a. Amber
 - b. Blue
 - c. Green
- 207.** Where is the centralised caution panel installed ●
- a. On the centre pedestal
 - b. At the pilots' eye level
 - c. On the overhead panel
- 208.** Where is the local warning light of a system covered by the centralised caution panel fitted ●
- a. On the glareshield
 - b. On the centralised caution panel
 - c. On the overhead panel
- 209.** What two forms of information do central warning displays give the flight crew ●
- a. Text and aural information
 - b. Graphic and tactile information
 - c. Text and graphic information
- 210.** What kind of system is ECAM ●
- a. A central warning system
 - b. A flight management system
 - c. An autoflight system
- 211.** Which two displays make up ECAM in the cockpit ●
- a. The primary flight display and the system display
 - b. The engine/warning display and the navigation display
 - c. The engine/warning display and the system display

- 212.** Apart from calling up pages and messages, which display property is set from the ECAM control panel ●
- a. The contrast ratio
 - b. The refresh rate
 - c. The brightness
- 213.** What is simulated when the TO CONFIG pushbutton is used ●
- a. The emergency electrical configuration
 - b. The approach configuration
 - c. The take-off power configuration
- 214.** Which two kinds of page does the system display show ●
- a. The system and status pages
 - b. The engine and warning pages
 - c. The memo and alert pages
- 215.** How many system pages exist for the system display ●
- a. Twelve
 - b. Eight
 - c. Five

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 216.** What happens to the related system when an illuminated pushbutton is pressed
- a. Its light test is only started
 - b. Its warning is only acknowledged
 - c. Its switching state is changed
- 217.** A pushbutton on the overhead panel has just lit up, so what does that tell the crew? ●
- a. The pushbutton lamp is being tested by the crew
 - b. The system has returned to its normal switching state
 - c. A local warning or an abnormal switching state exists
- 218.** Why are the pushbutton lights given a colour code ●
- a. To make abnormal conditions easy to identify
 - b. To show which bus supplies each panel
 - c. To group the pushbuttons by aircraft system
- 219.** An engine fire has broken out, so what colour will the associated pushbutton light show? ●
- a. Red
 - b. White
 - c. Amber
- 220.** Besides an abnormal pushbutton position, which condition is shown by a white light ●
- a. A backup system running
 - b. A failure needing awareness
 - c. A maintenance operation
- 221.** A pushbutton shows a steady blue legend, so what is the crew being told? ●
- a. A backup system is operating normally
 - b. A system has been left in an abnormal position
 - c. A temporarily used system is operating normally

- 222.** A pushbutton shows a steady green legend, so what is the crew being told? ●
- A temporary system is operating normally
 - A backup system is operating normally
 - A failure needing awareness has occurred
- 223.** Which shortcoming of the master caution and master warning lights led to a centralised panel ●
- They do not stay on long enough to be seen
 - They cannot be cancelled once they are on
 - They do not say which system is affected
- 224.** What does the centralised caution panel tell the crew about a fault ●
- Which particular system it is in
 - How many flight hours remain before rectification
 - Which circuit breaker has tripped
- 225.** How is the master caution light cancelled ●
- By switching off the affected system
 - By pressing its light cover
 - By pressing the local warning light
- 226.** Why does the crew cancel the master caution light rather than leave it on? ●
- So that the local warning light goes out
 - So that the centralised panel can be tested
 - So that another fault can be indicated
- 227.** What does the information on a central warning display describe ●
- The flight phase being computed
 - The present failure condition
 - The maintenance action already carried out
- 228.** An engineer compares the central warning systems of two different aircraft types, so what will he mostly find?
- The same function under different names
 - Different functions under the same name
 - The same name and the same hardware
- 229.** Under what condition is the symbol ADV displayed ●
- When a status page becomes available
 - When ECAM is working as a mono display
 - When a caution message is cancelled
- 230.** One of the two ECAM displays has failed, so which symbol appears in the lower part of the engine/warning display? ●
- STS
 - ADV
 - RCL
- 231.** Which two pages are combined in the cruise page ●
- The engine and fuel pages
 - The air and hydraulic pages
 - The engine and air pages

- 232.** A memo message has to be read, so which display and which part of it does the crew use? ●
- a. The lower part of the engine/warning display
 - b. The upper part of the engine/warning display
 - c. The lower part of the system display
- 233.** Which three things can be done to warnings and messages from the ECAM control panel ●
- a. Cancel, delete or call them up again
 - b. Dim, enlarge or colour them
 - c. Record, print or transmit them
- 234.** In which position must the slats and flaps be for the take-off configuration check to pass ●
- a. Fully extended
 - b. Fully retracted
 - c. In the T/O range
- 235.** What condition of the speed brakes is required by the take-off configuration check ●
- a. They must be retracted
 - b. They must be extended
 - c. They must be armed
- 236.** What condition of the wheel brakes is required by the take-off configuration check ●
- a. They must be released
 - b. They must be parked
 - c. They must not be overheated
- 237.** The take-off configuration check is run with a cabin door still open, so what will the crew find? ●
- a. The master caution will be inhibited
 - b. The normal message will not appear
 - c. The check will be bypassed automatically
- 238.** A continuous aural alert is sounding and the crew wants it silenced, so which ECAM control panel pushbutton is used? ●
- a. EMER CANCEL
 - b. CLR
 - c. RCL
- 239.** What happens when the ALL pushbutton is pressed and held ●
- a. The system pages are called up at one second intervals
 - b. The system pages are frozen on the display
 - c. The status page is called up again after five seconds
- 240.** What happens when the ALL pushbutton is pressed only briefly ●
- a. The displayed system page is deleted
 - b. The system page moves forward by one page
 - c. The system pages cycle continuously
- 241.** When does the light in the CLR pushbutton come on ●
- a. When warning, caution or status messages appear
 - b. When the recall function has just been used
 - c. When the system pages are being cycled by the crew

242. Which message is removed each time the CLR pushbutton is pushed ●
- The oldest warning in the list
 - The warning at the top
 - Every warning at once
243. What reappears on the display after the last clearance has been made with CLR ●
- The caution messages
 - The memo messages
 - The status page
244. How long after the status pushbutton is used do the normal messages return when no status messages are present ●
- After 5 seconds
 - After 10 seconds
 - After 2 seconds
245. The brightness of the engine/warning and system displays has to be changed, so which panel is used? ●
- The ECAM control panel
 - The centralised caution panel
 - The EICAS display select panel
246. How many system pages can be shown on the system display at any one time ●
- Four
 - Two
 - One
247. Which two air temperatures are shown among the permanent data on a system page ●
- TAT and SAT
 - TAT and EGT
 - SAT and ITT
248. Which item is shown as an alternative to the G-load among the permanent data ●
- The gross weight
 - The selected altitude
 - The static air temperature
249. A pilot needs the total air temperature while a system page is up, so where is it shown? ●
- On the engine/warning display
 - On the upper part of the page
 - On the lower part of the page

LEVEL 3 — APPLY AND REASON

250. A system is working normally but its pushbutton has been left out of its normal switching position, so what will the crew see on that pushbutton? ●
- It will flash until the fault is cleared
 - It will be lit even though no fault exists
 - It will stay dark because no fault exists

251. A pushbutton has stayed dark for the whole flight, so what does that say about its system? ●
- a. It is a backup that has taken over the load
 - b. It is normal and in its usual switching state
 - c. It has failed but needs no action at once
252. A pushbutton carries an amber FAULT legend, so how urgent is the crew action? ●
- a. Action is needed but not at once
 - b. Action is needed at once
 - c. Action is not needed at all
253. Several pushbuttons are lit at once and the crew must choose where to start, so which colour takes priority? ●
- a. Red
 - b. White
 - c. Amber
254. A pushbutton is lit in a colour that reports normal operation, so which two colours could it be? ●
- a. Red or amber
 - b. Amber or white
 - c. Green or blue
255. A system has been selected off and no fault exists, so which colour will its pushbutton show? ●
- a. Amber
 - b. Green
 - c. White
256. Which two colours on a pushbutton both report a failure of some kind? ●
- a. Red and amber
 - b. Green and blue
 - c. White and green
257. A fault occurs in a system whose local light is on the overhead panel out of the pilot's sight, so how does the crew learn which system it is? ●
- a. The master caution light names the affected system
 - b. The centralised caution panel names the affected system
 - c. The overhead local light is repeated on the glareshield
258. A fault is already being annunciated and a second fault then occurs, so what must the crew do for the new one to be indicated? ●
- a. Cancel the master caution light first
 - b. Reset the centralised caution panel first
 - c. Press the local warning light first
259. A crew is told that a fault exists but not which system has it, so which indication are they looking at? ●
- a. A central warning display message
 - b. A centralised caution panel light
 - c. A master caution light
260. A pilot sees STS in the lower part of the engine/warning display but no status page is being shown, so where will that status page appear once it is called up? ●
- a. On the engine/warning display
 - b. On the system display
 - c. On the centralised caution panel

- 261.** A page made up of the engine and air pages is on the system display, so where is the aircraft in its flight? ●
- In the take-off run
 - In cruise flight
 - In the approach
- 262.** A crew member has cleared several caution messages with CLR and then wants to read them again, so what must be done? ●
- Press CLR again to restore them
 - Press STS to bring up the status page
 - Press RCL to recall them
- 263.** The status pushbutton has been pressed and after a short time the normal messages return on their own, so what does that tell the crew? ●
- No status messages are present
 - The status page has been cleared by CLR
 - The ECAM is working as a mono display
- 264.** A level 3 alert has triggered an aural warning and the master warning lights, and the crew uses EMER CANC, so what is left showing on the ECAM? ●
- Nothing at all is left on the display
 - The warning message on the display stays
 - Only the system page is cancelled
- 265.** The light in the CLR pushbutton is on and nobody has pushed it yet, so what is on the ECAM display? ●
- A warning, caution or status message
 - Only the permanent data of the system page
 - Only the memo messages of the systems
- 266.** A pilot wants to step through every system page in turn, so what must be done with the ALL pushbutton? ●
- Press it twice in quick succession
 - Press it briefly once
 - Press and hold it
- 267.** STS is showing on the engine/warning display and the crew presses the status pushbutton, so what appears? ●
- The normal messages, since no status messages exist
 - The status page with its messages
 - The cruise page with its engine data
- 268.** A pilot wants to see a second system page while one is already up, so what happens to the page already displayed? ●
- It stays, because the second page opens on the E/WD
 - It is replaced, because only one can be shown at a time
 - It stays, because two pages can be shown side by side
- 269.** A pilot reading a system page wants the gross weight, so where on that page does he look? ●
- On the lower part, among the permanent data
 - On the upper part, among the engine parameters
 - On the right side, among the inoperative systems

Answers — Central warning systems

203. c All pushbuttons are dark during normal operation. (L1, PDF p. 859)
204. b Red is the colour used when an immediate action is necessary. (L1, PDF p. 859)
205. b Green indicates the normal operation of a backup system. (L1, PDF p. 860)
206. b Blue indicates the normal operation of a temporarily used system. (L1, PDF p. 860)
207. b The centralised caution panel is installed at the pilots' eye level. (L1, PDF p. 861)
208. c The local warning light sits on the overhead panel while the centralised panel sits in view. (L1, PDF p. 861)
209. c The displays provide text and graphic information. (L1, PDF p. 863)
210. a ECAM is a central warning system. (L1, PDF p. 865)
211. c ECAM is present as the E/WD and the SD. (L1, PDF p. 865)
212. c The control panel is used to control the brightness of the ECAM display units. (L1, PDF p. 867)
213. c The TO CONFIG pushbutton simulates the T/O power configuration. (L1, PDF p. 867)
214. a The SD shows the system and the status pages. (L1, PDF p. 871)
215. a One of twelve system pages can be shown. (L1, PDF p. 871)
216. c Pressing the illuminated pushbutton changes the state of the switch behind it. (L2, PDF p. 859)
217. c A pushbutton lights only for a local warning or for an abnormal switching state. (L2, PDF p. 859)
218. a The colour code exists to help in the easy identification of abnormal conditions. (L2, PDF p. 859)
219. a An engine fire needs immediate action, and immediate action is coded red. (L2, PDF p. 859)
220. c White covers a pushbutton in an abnormal position or a maintenance operation. (L2, PDF p. 860)
221. c Blue is reserved for the normal operation of a temporarily used system. (L2, PDF p. 860)
222. b Green is reserved for the normal operation of a backup system. (L2, PDF p. 860)
223. c The master lights give no information about which system is affected or which local light is on. (L2, PDF p. 861)
224. a The panel indicates a fault in a particular system. (L2, PDF p. 861)
225. b The master caution light is cancelled by pressing its light cover. (L2, PDF p. 861)
226. c Cancelling the light frees it so that a further fault can be indicated. (L2, PDF p. 861)
227. b The displays describe the present failure condition. (L2, PDF p. 863)
228. a Central warning systems in different types generally share the function but not the name. (L2, PDF p. 863)
229. b ADV appears when ECAM is working as a mono display, that is with only one display. (L2, PDF p. 865)
230. b With only one ECAM display left, ECAM works as a mono display and ADV is shown. (L2, PDF p. 865)
231. c The cruise page consists of the engine and air pages. (L2, PDF p. 865)
232. a Memo messages are in the lower part of the E/WD. (L2, PDF p. 865)
233. a The panel cancels, deletes or calls up again the warnings and messages. (L2, PDF p. 867)
234. c The check requires the slats and flaps to be in the T/O range. (L2, PDF p. 867)
235. a The check requires the speed brakes to be retracted. (L2, PDF p. 867)
236. c The check requires the wheel brakes not to be OVHT. (L2, PDF p. 867)
237. b Door closed is one of the items checked, so an open door prevents the normal result. (L2, PDF p. 867)
238. a The EMER CANC pushbutton cancels the aural warnings. (L2, PDF p. 867)
239. a Pressing and holding ALL calls up the system pages at one second intervals. (L2, PDF p. 867)
240. b A brief press moves the system page forward by one page. (L2, PDF p. 867)
241. a The CLR light comes on when warning, caution or status messages appear on the ECAM display. (L2, PDF p. 867)
242. b Each push of CLR deletes the warning at the top. (L2, PDF p. 867)
243. b After the last clearance the memo messages reappear. (L2, PDF p. 867)
244. a With no status messages present the normal messages are displayed again after 5 seconds. (L2, PDF p. 867)
245. a Brightness is controlled from the ECAM control panel. (L2, PDF p. 867)

- 246. c** Only one of the twelve system pages can be displayed at any one time. (L2, PDF p. 871)
- 247. a** The permanent data includes TAT and SAT. (L2, PDF p. 871)
- 248. b** The permanent data ends with G-load or selected altitude. (L2, PDF p. 871)
- 249. c** TAT is one of the permanent data items on the lower part of the system page. (L2, PDF p. 871)
- 250. b** A pushbutton lights for an abnormal switching state as well as for a local warning, so it is lit with no fault present. (L3, PDF p. 859)
- 251. b** A pushbutton lights only for a local warning or an abnormal switching state, so a dark one means neither is present. (L3, PDF p. 859)
- 252. a** Amber is used when action is required but not necessarily at once. (L3, PDF p. 859)
- 253. a** Red marks the failure needing immediate action while amber needs awareness but no immediate action. (L3, PDF p. 860)
- 254. c** Green and blue are the two colours printed against normal operation, of a backup system and of a temporarily used system. (L3, PDF p. 860)
- 255. c** A pushbutton left in an abnormal position is coded white, and no failure is being reported. (L3, PDF p. 860)
- 256. a** Red and amber are the two colours printed against a failure, the first needing immediate action and the second only awareness. (L3, PDF p. 860)
- 257. b** The master light says only that something is wrong, so the panel at eye level is what identifies the system. (L3, PDF p. 861)
- 258. a** Only after the master caution light has been cancelled can another fault be indicated. (L3, PDF p. 861)
- 259. c** The master lights give no information about the system affected, while the panel and the display both name it. (L3, PDF p. 863)
- 260. b** STS says a status page is available and the status page is displayed on the SD. (L3, PDF p. 865)
- 261. b** The cruise page, which consists of the engine and air pages, is the one shown in cruise flight. (L3, PDF p. 865)
- 262. c** CLR deletes the messages one by one and RCL is the pushbutton that recalls them. (L3, PDF p. 867)
- 263. a** The normal messages come back by themselves only when there are no status messages to show. (L3, PDF p. 867)
- 264. b** EMER CANC removes the aural warnings, the master warning lights and the caution messages, so a warning message itself is not in that list. (L3, PDF p. 867)
- 265. a** The CLR light comes on only when warning, caution or status messages are present. (L3, PDF p. 867)
- 266. c** Pressing and holding ALL calls up the system pages one after another at one second intervals. (L3, PDF p. 867)
- 267. b** STS means a status page is available, so pressing the pushbutton brings that page up rather than returning the normal messages. (L3, PDF p. 867)
- 268. b** Only one of the twelve system pages can be displayed at any one time. (L3, PDF p. 871)
- 269. a** GW is printed among the permanent data on the lower part of the system page. (L3, PDF p. 871)

Alert levels, colours and flight phases

PDF pages 877–893 · 57 questions

LEVEL 1 — RECALL

- 270.** Which aircraft condition is associated with a level 3 alert ●
- a. A configuration to monitor
 - b. An abnormal configuration
 - c. An emergency configuration
- 271.** Which aircraft condition is associated with a level 2 alert ●
- a. An abnormal configuration
 - b. An emergency configuration
 - c. A configuration to monitor
- 272.** What priority do level 3 warnings carry ●
- a. A medium priority
 - b. The highest priority
 - c. The lowest priority
- 273.** How quickly must the crew react to a level 3 warning ●
- a. Before the next scheduled check
 - b. Before the next flight phase
 - c. Immediately
- 274.** What influence does a level 2 caution have on the flight ●
- a. A direct and immediate influence
 - b. No direct influence
 - c. An influence only during the approach
- 275.** What is required of the crew when a level 2 caution appears ●
- a. An immediate reaction
 - b. Greater attention
 - c. No reaction at all
- 276.** On which display do level 1 cautions appear ●
- a. The system display
 - b. The engine/warning display
 - c. The centralised caution panel
- 277.** Which indications are absent for a level 1 caution ●
- a. Memo messages and status messages
 - b. Aural alarms and warning lights
 - c. Text messages and system pages
- 278.** Which flight phase does the flight warning system select by default when it has no data ●
- a. The preflight phase
 - b. The approach phase
 - c. The cruise phase

- 279.** Which parameters are permanently displayed on the upper EICAS ●
- The status information
 - The primary engine parameters
 - The secondary engine parameters
- 280.** How many EFIS/EICAS interface units are fitted ●
- Four
 - Three
 - Two
- 281.** Which two parameters are the primary engine parameters on the EICAS display ●
- N2 rotor speed and fuel flow in kilogrammes
 - N1 rotor speed and exhaust gas temperature
 - Oil pressure and oil temperature in degrees
- 282.** Which colour is used for a Level A warning message in the alert message field ●
- White
 - Red
 - Amber
- 283.** How many manual snapshots can be stored for each system ●
- Up to 3
 - Up to 5
 - Up to 10
- 284.** A snapshot has just been taken because a parameter went outside its preset value, so where is it kept? ●
- In the manual event non-volatile memory
 - In the auto event non-volatile memory
 - In the real time data storage buffer

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 285.** Which alert level is classified as a warning rather than a caution ●
- Level 2
 - Level 1
 - Level 3
- 286.** An alert is classified as a level 3 rather than a level 2, so what does that say about the aircraft? ●
- There is a real emergency on board
 - There is no direct influence on the flight
 - There is a system that is not needed
- 287.** The aircraft has gone into an abnormal configuration without any emergency, so which alert level is produced? ●
- Level 3
 - Level 2
 - Level 1
- 288.** In what manner does the master warning light behave for a level 3 warning ●
- It stays dark
 - It flashes
 - It stays steady

- 289.** What may appear on the system display in addition to a level 3 warning ●
- a. A system page, if one is available
 - b. A cruise page, if one is available
 - c. A status page, if one is available
- 290.** An alarm sounds and the master warning light begins to flash, so which level has been reached? ●
- a. Level 3
 - b. Level 2
 - c. Level 1
- 291.** How many chimes are sounded for a level 2 caution ●
- a. One
 - b. None
 - c. Three
- 292.** What kind of system does a level 1 caution draw attention to ●
- a. A serviceable system temporarily switched off
 - b. A defective system needed for the flight to continue
 - c. A defective system not required for the flight to continue
- 293.** A caution message has appeared on the engine/warning display with no sound and no light, so which level has been reached? ●
- a. Level 3
 - b. Level 1
 - c. Level 2
- 294.** A brakes channel fault is annunciated as a level 1 caution, so what does that say about continuing the flight? ●
- a. That system is required to continue
 - b. The flight has to be ended at once
 - c. That system is not required to continue
- 295.** From which system does the flight warning system take its compressed or not compressed signal
- a. The landing gear system
 - b. The air data system
 - c. The brake control system
- 296.** Which number is given to the default flight phase of the flight warning system ●
- a. 12
 - b. 1
 - c. 8
- 297.** Which four items can be shown on the lower EICAS ●
- a. Primary engine parameters, synoptics, memo messages and maintenance pages
 - b. Secondary engine parameters, system pages, status information and alert messages
 - c. Secondary engine parameters, synoptics, status information and maintenance pages
- 298.** Which panel brings synoptics and status information onto the lower display unit ●
- a. The ECAM control panel
 - b. The EICAS display select panel
 - c. The centralised caution panel

- 299.** Through which unit are the maintenance pages called up on the lower EICAS ●
- The CMC
 - The EDSP
 - The EIU
- 300.** Which data appears on the lower EICAS immediately after power-up ●
- The status information
 - The secondary engine data
 - The primary engine data
- 301.** Which data bus supplies the EICAS display units ●
- An ARINC 573 data bus
 - An ARINC 629 data bus
 - An ARINC 429 data bus
- 302.** A message shows in the alert message field in amber and indented, so which level is it? ●
- Level D memo
 - Level B caution
 - Level C advisory
- 303.** Why is information that is not required removed from the EICAS display ●
- To reduce the heat produced by the screens
 - To save electrical power in the display units
 - To maintain a quiet, dark flight deck
- 304.** What makes an automatic snapshot occur for a system ●
- Parameters that exceed preset values
 - Selection of the real time display mode
 - A request made by the maintenance crew
- 305.** What causes a manual snapshot of the real time format to be taken ●
- An action by the flight or maintenance crew
 - A report sent to the printer or to ACARS
 - A parameter that goes out of its limits

LEVEL 3 — APPLY AND REASON

- 306.** Two alerts come up together, the first at level 3 and the second at level 1, so what does the more serious of the two report? ●
- A configuration to monitor
 - An emergency configuration
 - An abnormal configuration
- 307.** The cabin altitude has become excessive, so what indications will the crew get? ●
- A single chime with a steady master caution light
 - A message on the display with no aural alarm
 - An aural alarm with a flashing master warning light
- 308.** An engine fire breaks out in flight, so what must the crew do about it? ●
- React immediately, as it is a real emergency
 - Note it and continue, as the flight is not affected
 - React before the next flight phase begins

- 309.** A level 3 warning and a level 1 caution are annunciated at the same moment, so which one is handled first? ●
- Either one, since the two carry the same priority
 - The warning, which has the highest priority
 - The caution, which has the highest priority
- 310.** A single chime sounds and the master caution light comes on, so what does the crew know about the failure? ●
- It has no direct influence on the flight
 - It gives no aural alarm and no warning light
 - It is a real emergency needing an immediate reaction
- 311.** A failure needs greater attention but no immediate reaction, so which aural alert will the crew hear? ●
- No sound at all
 - A single chime
 - A continuous alarm
- 312.** A defective system is annunciated with a single chime rather than silently, so what does the crew know about it? ●
- It is a level 2, needing greater attention
 - It is a level 1, needing greater attention
 - It is a level 3, needing an immediate reaction
- 313.** A system that the flight does not need has become defective, so what will the crew hear? ●
- Nothing, as there is no aural alarm
 - A continuous alarm from the master warning
 - A single chime from the master caution
- 314.** An alert needs the crew to read the engine/warning display to notice it at all, so what does that say about its level? ●
- It is a level 2, which gives only a single chime
 - It is a level 3, which gives only a flashing light
 - It is a level 1, which has no attention getter
- 315.** A recorder fault and a stall warning occur in the same flight, so how do the two alerts differ in what the crew gets? ●
- One gives no sound or light while the other gives both
 - One gives a single chime while the other gives no sound
 - Each one gives an aural alarm but only one gives a light
- 316.** An instructor asks what separates a warning from a caution in practice, so what is the difference in the reaction required? ●
- A caution needs an immediate reaction, a warning does not
 - A warning needs an immediate reaction, a caution does not
 - Each one needs the same immediate reaction from the crew
- 317.** Which indication is common to all three alert levels when it is available ●
- A chime sounding from the loudspeaker
 - A light coming on at the eye level
 - A system page on the system display

- 318.** A technician wants to bring the maintenance pages up on the lower EICAS, so why will the display select panel not do it? ●
- Because that panel only works after the aircraft is powered up
 - Because that panel only selects the primary engine parameters
 - Because that panel only selects synoptics and status information
- 319.** An aircraft has just been powered up and no selection has been made, so what is seen on the two EICAS display units? ●
- Primary engine data above and secondary engine data below
 - Primary engine data above and the status information below
 - Secondary engine data above and primary engine data below
- 320.** All three interface units have failed, so what will the EICAS display units show? ●
- Nothing, as their data comes through them
 - Primary engine data only, taken direct from the engines
 - Maintenance pages only, taken direct from the CMC
- 321.** A pilot needs the secondary engine parameters while a synoptic is up on the lower EICAS, so what must happen on that display? ●
- The synoptic must be deselected to show them
 - The synoptic and the parameters appear together
 - The upper display must be used to show them
- 322.** A synoptic is wanted on the lower EICAS straight after power-up, so what must be done? ●
- It must be selected on the display select panel
 - It is already there after power-up
 - It must be called up through the maintenance computer
- 323.** The upper EICAS display unit is showing data that cannot be deselected, so what is on it? ●
- The synoptics of the systems
 - The primary engine parameters
 - The secondary engine parameters
- 324.** Every indication that is not needed has been removed from the main EICAS display, so which two groups are still shown? ●
- The landing gear status and the flap indications
 - The cabin altitude and the duct pressure readouts
 - The primary engine parameters and the fuel system data
- 325.** A caution and an advisory both show in amber in the alert message field, so how is the advisory told apart? ●
- It is underlined
 - It is flashing
 - It is indented
- 326.** An engine exceedance occurs while the aeroplane stands with its parking brake set, so what becomes of the event? ●
- It is lost, since two engines have to be running
 - It is stored, since that storage is not inhibited
 - It is lost, since the brake has to be released first

- 270. c** Level 3 is the warning raised for an emergency configuration. (L1, PDF p. 878)
- 271. a** Level 2 is the caution raised for an abnormal configuration. (L1, PDF p. 878)
- 272. b** Level 3 warnings are the warnings with the highest priority. (L1, PDF p. 879)
- 273. c** A level 3 warning is a real emergency and the crew must react immediately. (L1, PDF p. 879)
- 274. b** Level 2 cautions have no direct influence on the flight. (L1, PDF p. 881)
- 275. b** A level 2 caution calls for greater attention. (L1, PDF p. 881)
- 276. b** Level 1 cautions appear on the E/WD. (L1, PDF p. 883)
- 277. b** There are no aural alarms or warning lights for a level 1 caution. (L1, PDF p. 883)
- 278. c** With the data unavailable the FWS defaults to the cruise phase. (L1, PDF p. 885)
- 279. b** The upper EICAS permanently displays the primary engine parameters. (L1, PDF p. 887)
- 280. b** There are three EFIS/EICAS interface units. (L1, PDF p. 887)
- 281. b** The page names N1 rotor speed and EGT as the primary engine parameters. (L1, PDF p. 889)
- 282. b** Level A warning messages are printed as showing in red. (L1, PDF p. 889)
- 283. b** Storage is available for up to 5 manual snapshots per system. (L1, PDF p. 893)
- 284. b** Auto snapshots go to the auto event non-volatile memory. (L1, PDF p. 893)
- 285. c** Level 3 carries the title warnings while levels 2 and 1 are cautions. (L2, PDF p. 877)
- 286. a** Level 3 is printed against a real emergency, while a level 2 is a failure with no direct influence on the flight. (L2, PDF p. 877)
- 287. b** An abnormal configuration is the condition behind a level 2 caution. (L2, PDF p. 878)
- 288. b** The master warning light flashes for a level 3 warning. (L2, PDF p. 879)
- 289. a** If available, a system page may also appear on the SD. (L2, PDF p. 879)
- 290. a** An aural alarm with a flashing master warning light is the level 3 signature. (L2, PDF p. 879)
- 291. a** A level 2 caution is given with a single chime. (L2, PDF p. 881)
- 292. c** Level 1 cautions point at a defective system which is not required for the flight to continue. (L2, PDF p. 883)
- 293. b** Level 1 is the only level with no aural alarm and no warning light. (L2, PDF p. 883)
- 294. c** Level 1 cautions cover a defective system which is not required for the flight to continue. (L2, PDF p. 883)
- 295. a** The gear compressed or not compressed information comes from the L/G system. (L2, PDF p. 885)
- 296. c** The default is the cruise phase, numbered 8. (L2, PDF p. 885)
- 297. c** The lower EICAS can show secondary engine parameters, synoptics, status information and maintenance pages. (L2, PDF p. 887)
- 298. b** Synoptics and status information are selected with the EICAS display select panel (EDSP). (L2, PDF p. 887)
- 299. a** The maintenance pages are called up via the CMC. (L2, PDF p. 887)
- 300. b** After power-up the lower EICAS shows the secondary engine data. (L2, PDF p. 887)
- 301. c** The display units are supplied via an ARINC 429 data bus. (L2, PDF p. 887)
- 302. c** Only the Level C advisory messages are both amber and indented. (L2, PDF p. 889)
- 303. c** Removing unneeded information is what keeps the flight deck quiet and dark. (L2, PDF p. 889)
- 304. a** An auto snapshot is taken when parameters go beyond their preset values. (L2, PDF p. 893)
- 305. a** A manual snapshot follows an action by the flight or maintenance crew. (L2, PDF p. 893)
- 306. b** Level 3 is the warning raised by an emergency configuration, while level 1 only covers a configuration to monitor. (L3, PDF p. 878)
- 307. c** Excess cabin altitude is a level 3 warning and level 3 gives an aural alarm with a flashing master warning light. (L3, PDF p. 879)
- 308. a** Engine fire is a level 3 warning and a level 3 warning is a real emergency needing an immediate reaction. (L3, PDF p. 879)
- 309. b** Level 3 warnings are the warnings with the highest priority. (L3, PDF p. 879)
- 310. a** A single chime with the master caution light is a level 2, and level 2 failures have no direct influence on the flight. (L3, PDF p. 881)

- 311. b** Greater attention without an immediate reaction is a level 2, and a level 2 is given with a single chime. (L3, PDF p. 881)
- 312. a** A single chime marks a level 2, which calls for greater attention, while a level 1 would have been silent. (L3, PDF p. 881)
- 313. a** A defective system not required to continue the flight is a level 1 caution, and level 1 has no aural alarm. (L3, PDF p. 883)
- 314. c** Only a level 1 caution comes without an aural alarm or a warning light, so nothing draws the eye to it. (L3, PDF p. 883)
- 315. a** A recorder fault is a level 1 with no aural alarm or warning light, while a stall warning is a level 3 with an aural alarm and a flashing master warning light. (L3, PDF p. 883)
- 316. b** The level 3 warning is a real emergency needing an immediate reaction, while the level 2 caution has no direct influence on the flight. (L3, PDF p. 883)
- 317. c** Each of the three levels is printed with the note that a system page may also appear on the SD if available. (L3, PDF p. 883)
- 318. c** The EDSP selects synoptics and status information while the maintenance pages come via the CMC. (L3, PDF p. 887)
- 319. a** After power-up the upper unit shows the primary and the lower unit the secondary engine data. (L3, PDF p. 887)
- 320. a** The display units are fed by one of the three EIUs, and all the important systems supply their data to those units. (L3, PDF p. 887)
- 321. a** The lower EICAS carries the secondary parameters, synoptics, status and maintenance pages in turn, while the upper one is permanently the primary parameters. (L3, PDF p. 887)
- 322. a** After power-up the lower unit shows the secondary engine data, so a synoptic has to be selected on the EDSP. (L3, PDF p. 887)
- 323. b** The upper EICAS permanently displays the primary engine parameters, the selectable items being on the lower unit. (L3, PDF p. 887)
- 324. c** Those two groups are the ones stated to show at all times. (L3, PDF p. 889)
- 325. c** The advisory is the amber message that is indented. (L3, PDF p. 889)
- 326. b** Engine exceedance page storage is printed as not inhibited, so the running and brake conditions do not apply to it. (L3, PDF p. 893)

Ground proximity warning principles

PDF pages 897–905 • 32 questions

LEVEL 1 — RECALL

- 327.** What is the purpose of the GPWS ●
- To warn the flight crew of another aircraft on a conflicting track
 - To warn the flight crew when the aircraft comes too close to the ground
 - To warn the flight crew of a windshear encountered on approach
- 328.** By which two means is the basic GPWS warning given to the pilot ●
- A master caution and a single chime
 - A flashing display and a stick shaker
 - An aural alert and annunciator lights
- 329.** On which display does the enhanced ground proximity warning system show the surrounding terrain ●
- On the primary flight display
 - On the navigation display
 - On the engine indication display
- 330.** Which three parts are the main parts of a ground proximity warning system ●
- One warning computer, indications and inhibit switches
 - One warning computer, a radio altimeter and a speaker
 - Two warning computers, indications and a control panel
- 331.** A basic GPWS is checked, so at which radio altitude does its operative range begin? ●
- 30 ft
 - 50 ft
 - 10 ft
- 332.** An aircraft is fitted with the enhanced system, so which two functions does it have on top of the basic modes? ●
- Radio altitude callouts, and the modulation of alerting envelopes
 - Terrain awareness alerting and display, and terrain clearance floor
 - Windshear alerting for take-off, and excessive bank angle alerting
- 333.** The enhanced computer needs the height above the ground for its warning calculation, so which unit supplies it? ●
- The digital air data computer
 - The instrument landing system
 - The downward looking radio altimeter

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 334.** A ground proximity alert sounds in the flight deck, so what kind of voice is heard? ●
- A rising tone sequence
 - A recorded crew voice
 - A synthetic voice

- 335.** An operator asks where a ground proximity warning system is only rarely found, so which aircraft are meant? ●
- a. Twin-engined aeroplanes
 - b. Helicopters
 - c. Four-engined aeroplanes
- 336.** Where else are GPWS alert messages presented in some aircraft ●
- a. On the primary flight displays
 - b. On the navigation displays
 - c. On the engine indication display
- 337.** In which aeroplanes is the ground proximity warning system installed ●
- a. In all of them except the very small ones
 - b. In the very small ones only
 - c. In all of them without any exception
- 338.** Which signal are the terrain functions of the enhanced system independent of ●
- a. The radio altitude
 - b. The barometric altitude
 - c. The glide slope deviation
- 339.** A crew is told their fit is the advanced version, so what does it do that the basic one does not? ●
- a. It sounds its alerts with a synthetic voice
 - b. It shows the terrain around the aircraft
 - c. It lights annunciators in front of the pilot
- 340.** From which sources does the GPWC receive its inputs ●
- a. From an air data computer and various navigation systems
 - b. From the autopilot and the flight control computers
 - c. From the flight data recorder and the engine computers
- 341.** Only some aeroplanes carry one of the three main parts of the system, so which part is it? ●
- a. The warning computer
 - b. The indications
 - c. The inhibit switches
- 342.** Which kind of accident is the enhanced system intended to prevent ●
- a. Runway incursion on landing
 - b. Loss of control in flight
 - c. Controlled flight into terrain
- 343.** With what does the enhanced system compare the aircraft position ●
- a. With an internal database
 - b. With the radio altitude signal
 - c. With the weather radar picture
- 344.** Which two sources can supply the aircraft position used in that comparison ●
- a. The ADC or the IRS
 - b. The FMS or the GPS
 - c. The ILS or the VOR

- 345.** Windshear alerting is fitted to an aircraft, so which mode number does it carry? ●
- Mode 6
 - Mode 7
 - Mode 5
- 346.** What does the enhanced system do with the variety of aircraft parameters it accepts ●
- It transmits them to the ground station
 - It records them for later readout
 - It applies alerting algorithms to them
- 347.** Which two products were introduced in the late 1980s
- MARK VI for digital aircraft and MARK I for analogue aircraft
 - MARK V for digital aircraft and MARK VII for analogue aircraft
 - MARK II for analogue aircraft and MARK III for digital aircraft
- 348.** Which two introductions improve the terrain awareness and the warning times of the enhanced system ●
- The terrain display and the terrain data base look ahead protection
 - The terrain clearance floor and the modulation of the envelopes
 - The altitude callouts and the alert for an excessive bank angle
- 349.** Why can the warning time be short when the aircraft flies towards steep terrain ●
- The aural message volume is reduced to one half below 300 feet
 - The primary sensor is a downward looking radio altimeter
 - The alerting boundaries are computed from the barometric height

LEVEL 3 — APPLY AND REASON

- 350.** Rising ground is approached at 2600 ft radio altitude, so why does the basic GPWS stay quiet? ●
- Its boundaries are computed from the barometric altitude
 - Its automatic operation begins only at a lower radio altitude
 - Its aural alerts are suppressed with the landing gear up
- 351.** The radio altimeter has failed in flight, so what can still warn the crew about high ground ahead? ●
- The enhanced system terrain functions
 - The below glide slope alert of mode 5
 - The basic ground proximity modes
- 352.** The crew sees the surrounding high ground pictured on the navigation display at 3000 ft, so which system provides it? ●
- The weather radar system
 - The enhanced system
 - The basic warning system
- 353.** Which GPWS function keeps working when the radio altitude signal is no longer available ●
- The terrain awareness alerting and display
 - The unsafe terrain clearance mode
 - The excessive terrain closure rate mode

- 354.** The limits of one of the alerting envelopes are exceeded, so what does the enhanced system give the flight crew? ●
- An automatic pull-up commanded through the autopilot
 - Aural alert messages with visual annunciations and displays
 - A master warning that has to be reset by the crew
- 355.** A stable non-precision approach is flown in low visibility with the gear down and the landing flaps out, so why does the early system stay silent?
- It is inhibited below 500 feet
 - It expects the aeroplane to land
 - It has lost its radio altitude input
- 356.** A system is required to see steep rising ground that the downward looking sensor would find too late, so which protection is needed? ●
- The modulation of the alerting envelopes
 - The terrain clearance floor protection
 - The terrain data base look ahead protection
- 357.** The aircraft is recognised as landing at one of the troublesome airports held in the database, so what does the system do?
- It adjusts the warning modes for that airport
 - It changes over to the terrain display only
 - It switches off all of its aural warnings
- 358.** A terrain clearance floor function is found on an aircraft, so which generation of the system is installed? ●
- The enhanced system
 - The MARK I system
 - The MARK V system

Answers — Ground proximity warning principles

- 327. b** The system warns the crew when the aircraft mistakenly comes too close to the ground. (L1, PDF p. 897)
- 328. c** The warning is an aural alert plus annunciator lights in front of the pilot. (L1, PDF p. 897)
- 329. b** The terrain picture is presented on the navigation display. (L1, PDF p. 899)
- 330. a** The computer, the indications and the inhibit switches are the main parts. (L1, PDF p. 901)
- 331. a** The printed operative range of the basic GPWS begins at 30 ft radio altitude. (L1, PDF p. 903)
- 332. b** Those two are the additional functions listed for the enhanced system. (L1, PDF p. 903)
- 333. c** The downward looking radio altimeter is the primary sensor for the warning calculation. (L1, PDF p. 905)
- 334. c** The aural alert uses a synthetic voice. (L2, PDF p. 897)
- 335. b** The page states the system is rarely installed in helicopters. (L2, PDF p. 897)
- 336. a** Some aircraft add alert messages on the primary flight displays. (L2, PDF p. 897)
- 337. a** Every aeroplane except the very small ones carries the system. (L2, PDF p. 897)
- 338. a** The enhanced terrain functions do not depend on the radio altitude. (L2, PDF p. 899)
- 339. b** The terrain picture is what the advanced system adds. (L2, PDF p. 899)
- 340. a** Its inputs come from an air data computer and from the navigation systems. (L2, PDF p. 901)
- 341. c** The inhibit switches are the part fitted to some aeroplanes only. (L2, PDF p. 901)
- 342. c** Its stated purpose is to help prevent CFIT accidents. (L2, PDF p. 903)
- 343. a** The aircraft position is compared with an internal database. (L2, PDF p. 903)
- 344. b** The position comes from the FMS or the GPS. (L2, PDF p. 903)

- 345. b** Windshear alerting is mode 7. (L2, PDF p. 903)
- 346. c** The parameters are taken in and alerting algorithms are applied to them. (L2, PDF p. 903)
- 347. b** The two late 1980s products were MARK V for digital and MARK VII for analogue aircraft. (L2, PDF p. 905)
- 348. a** Terrain display and terrain data base look ahead protection are the named improvements. (L2, PDF p. 905)
- 349. b** The warning is calculated from a downward looking sensor, which sees steep terrain late. (L2, PDF p. 905)
- 350. b** The aircraft is still above the radio altitude at which the basic system begins to operate automatically. (L3, PDF p. 897)
- 351. a** The enhanced terrain functions work independently of the radio altitude, while the basic modes need it. (L3, PDF p. 899)
- 352. b** The basic system works only below 2500 ft, while the enhanced terrain picture is independent of the radio altitude. (L3, PDF p. 899)
- 353. a** The enhanced terrain functions are independent of the radio altitude while the basic modes need it. (L3, PDF p. 899)
- 354. b** Exceeding an envelope limit produces aural messages together with visual annunciations and displays. (L3, PDF p. 903)
- 355. b** In the landing configuration the early system expects a landing and issues no warning. (L3, PDF p. 905)
- 356. c** Look ahead protection works from the database instead of the downward looking sensor. (L3, PDF p. 905)
- 357. a** At such an airport the warning modes are adjusted, and only under specific conditions. (L3, PDF p. 905)
- 358. a** The terrain clearance floor belongs to the enhanced system only. (L3, PDF p. 905)

GPWS modes 1 to 5

PDF pages 907–917 · 59 questions

LEVEL 1 — RECALL

- 359.** Which unit is the control unit of the EGPWS ●
- a. The air data computer
 - b. The display management computer
 - c. The enhanced computer
- 360.** Which condition does mode 1 alert on ●
- a. An excessive terrain closure rate
 - b. An altitude loss after take-off
 - c. An excessive descent rate
- 361.** Which aural message is generated when the aircraft penetrates the outer alert boundary of mode 1 ●
- a. PULL UP
 - b. DON'T SINK
 - c. SINK RATE
- 362.** Which condition does mode 2 alert on ●
- a. An unsafe clearance from the terrain
 - b. An excessive terrain closure rate
 - c. An excessive descent rate
- 363.** For how long after take-off is the mode 2B configuration valid ●
- a. For the first 60 seconds
 - b. For the first 90 seconds
 - c. For the first 30 seconds
- 364.** Which condition does mode 3 alert on ●
- a. An excessive rate of descent
 - b. An altitude loss after take-off
 - c. An unsafe clearance from terrain
- 365.** A DON'T SINK message is heard in the flight deck, so which mode has generated it? ●
- a. Mode 4
 - b. Mode 3
 - c. Mode 1
- 366.** Which condition does mode 4 alert on ●
- a. An excessive rate of descent
 - b. An unsafe clearance from the terrain
 - c. An altitude loss after the take-off
- 367.** How many submodes does mode 4 have ●
- a. Three
 - b. Four
 - c. Two

- 368.** Which condition does mode 5 alert on ●
- a. An unsafe clearance from the terrain
 - b. An excessive rate of descent near ground
 - c. A deviation below the glide slope beam
- 369.** How many levels of mode 5 operation does the computer provide ●
- a. One
 - b. Three
 - c. Two

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 370.** Which additional protection does mode 6 provide
- a. An alert for an excessive rate of closure with the terrain
 - b. A comparison of the aircraft position with a terrain database
 - c. A selectable menu of radio altitude callouts during the approach
- 371.** Which part of the system operations is independent of the other functions ●
- a. The built-in test equipment function
 - b. The terrain awareness display function
 - c. The warning function
- 372.** The descent continues after the first mode 1 message and the inner alert boundary is penetrated, so what is heard then? ●
- a. PULL UP
 - b. TERRAIN TERRAIN
 - c. TOO LOW TERRAIN
- 373.** Which three inputs define the two alert boundaries of mode 1 ●
- a. Barometric altitude, glide slope deviation and gear position
 - b. Radio altitude, computed airspeed and the position of the flaps
 - c. Barometric vertical speed, inertial vertical speed and radio altitude
- 374.** What does the computer produce besides the aural message in both mode 1 cases ●
- a. A discrete output to drive the visual annunciations
 - b. A terrain picture on the navigation display
 - c. A command to the autopilot to level the aircraft
- 375.** In which flight conditions can the aircraft be when a mode 2 alert is generated ●
- a. In descent or level flight only
 - b. In the landing configuration only
 - c. In descent, level flight or climb
- 376.** A mode 2 alert boundary is being computed, so which two signals are combined? ●
- a. Barometric altitude and glide slope deviation
 - b. Airspeed and the position of the flaps
 - c. Vertical speed and radio altitude
- 377.** When is the mode 2A alert generated ●
- a. With the flaps set to the landing position on an approach
 - b. In any configuration other than the one defined for 2B
 - c. Within the first sixty seconds following the take-off

- 378.** Which aural message comes first when the aircraft penetrates the mode 2A alerting boundaries ●
- a. SINK RATE
 - b. TERRAIN TERRAIN
 - c. PULL UP
- 379.** The aircraft goes on penetrating the mode 2A boundaries after the first message, so which message is repeated until it leaves the envelope? ●
- a. TERRAIN TERRAIN
 - b. PULL UP
 - c. DON'T SINK
- 380.** What makes the upper alert boundary of mode 2A vary ●
- a. The rate of closure with the terrain
 - b. The speed of the aircraft
 - c. The position of the landing flaps
- 381.** Within how many dots of the localizer or glide slope centreline does the mode 2B configuration apply during an ILS approach ●
- a. Within plus or minus 3 dots
 - b. Within plus or minus 1 dot
 - c. Within plus or minus 2 dots
- 382.** Which messages are suppressed when both gear and flaps are in the landing position in mode 2B ●
- a. The TERRAIN messages
 - b. The PULL UP messages
 - c. The TOO LOW GEAR messages
- 383.** In which two flight phases does mode 3 provide its alerts ●
- a. During the cruise and during a precision approach
 - b. Immediately after take-off and during a missed approach
 - c. During the final approach and the landing roll
- 384.** Until when do the mode 3 visual annunciators remain active ●
- a. Until a positive rate of climb is established
 - b. Until the warning envelope has been left
 - c. Until the landing gear has been retracted
- 385.** Which configuration must exist after take-off before the computer initiates a mode 3 alert
- a. The gear up and the flaps not at landing
 - b. The gear down and the flaps at landing
 - c. The gear up and the flaps set to landing
- 386.** With reference to what is the insufficient terrain clearance of mode 4 detected ●
- a. The phase of flight and the speed
 - b. The glide slope and the localizer
 - c. The vertical speed and the bank angle
- 387.** When is mode 4A active ●
- a. During take-off with the gear and the flaps still retracted
 - b. During cruise and approach with the landing gear not at landing
 - c. During an ILS approach with the flaps in landing position

- 388.** At which speed does a mode 4A penetration with the landing gear up give the TOO LOW GEAR message ●
- At 159 knots or below
 - At 250 knots and above
 - At 190 knots or below
- 389.** The airspeed builds up during the cruise, so what happens to the mode 4A upper boundary? ●
- It decreases linearly
 - It increases linearly
 - It stays constant
- 390.** In which configuration does mode 4B give the TOO LOW FLAPS message ●
- With the gear up below one hundred and fifty nine knots
 - With the gear and the flaps both still fully retracted
 - With the gear down and the flaps not at landing
- 391.** On what is mode 4C based ●
- A minimum terrain clearance floor that increases during take-off
 - A deviation of the flight path below the glide slope beam
 - A fixed boundary of five hundred feet during the approach
- 392.** An ILS approach is being flown, so on which course does mode 5 give its alerts? ●
- A front course ILS approach
 - A back course ILS approach
 - A non-precision radio approach
- 393.** How loud is the soft alert GLIDE SLOPE message ●
- About one half of the other alert volume levels
 - About twice the other alert message volume levels
 - The same as the other alert message volume levels
- 394.** A hard alert activation occurs on an approach, so below which radio altitude is the aircraft? ●
- 500 feet
 - 300 feet
 - 245 feet
- 395.** How far below the glide slope must the aircraft be for the hard alert activation ●
- More than 2 dots
 - More than 3 dots
 - More than 1.3 dots

LEVEL 3 — APPLY AND REASON

- 396.** The terrain awareness display function has been lost, so what happens to the ground proximity warning functions? ●
- They are lost together with the display function
 - They fall back to the excessive descent rate mode
 - They go on working while the necessary inputs are there

- 397.** An installation carries only the modes that are basic system requirements, so which basic GPWS alerting mode is missing? ●
- Mode 4
 - Mode 6
 - Mode 5
- 398.** A warning lamp stays dark although the aural message is heard, so which computer function is suspect?
- The alert output processing
 - The terrain awareness display
 - The input signal processing
- 399.** A replacement enhanced computer is being fitted, so what has to be set so that it works with the right modes?
- The program pins, before it is installed
 - The maintenance pages, after the installation
 - The inhibit switches, before the first flight
- 400.** The crew reduces the descent rate as soon as the first mode 1 message is heard, so which boundary has not been penetrated? ●
- The inner alert boundary
 - The outer alert boundary
 - The upper alert boundary
- 401.** An aircraft descends rapidly towards flat ground with the gear up, so which mode produces the first alert? ●
- Mode 3
 - Mode 1
 - Mode 5
- 402.** Both the landing gear and the flaps are in the landing position when the mode 2B envelope is penetrated, so which message is repeated? ●
- TERRAIN
 - TERRAIN TERRAIN
 - PULL UP
- 403.** The mode 2B envelope is penetrated with the landing gear down but the flaps still up, so which message is heard first? ●
- PULL UP
 - TOO LOW TERRAIN
 - TERRAIN TERRAIN
- 404.** The flaps are at the landing position and the aircraft is within two dots of the localizer on an ILS approach, so which submode boundary applies? ●
- Mode 4B
 - Mode 2A
 - Mode 2B
- 405.** Which signal is used for the alert boundaries of both the excessive descent rate mode and the excessive terrain closure rate mode besides the radio altitude ●
- The glide slope deviation
 - The computed airspeed
 - The vertical speed

406. Which aural message follows a continued penetration in both the excessive descent rate mode and the excessive terrain closure rate mode ●
- a. TERRAIN TERRAIN
 - b. SINK RATE
 - c. PULL UP
407. A go-around is flown with the gear up and the flaps away from landing when a larger altitude loss than permitted is detected, so which alert is initiated? ●
- a. The mode 3 alert
 - b. The mode 1 alert
 - c. The mode 4 alert
408. A computer has to decide how much height may be lost before a mode 3 alert is given, so what does it do?
- a. It calculates the amount of permitted loss
 - b. It compares the loss with a fixed boundary
 - c. It reads the loss from the terrain database
409. Which mode works from the altitude above sea level rather than from a terrain closure rate ●
- a. Mode 2
 - b. Mode 3
 - c. Mode 1
410. Climbing out with gear and flaps up the aircraft loses more height than permitted while staying above the mode 4C filter value, so which message is heard? ●
- a. DON'T SINK
 - b. TOO LOW TERRAIN
 - c. SINK RATE
411. The mode 4A boundary is penetrated at 250 knots with the landing gear still retracted, so which message does the computer generate? ●
- a. TOO LOW TERRAIN
 - b. TOO LOW GEAR
 - c. TOO LOW FLAPS
412. Just after take-off with gear and flaps up the radio altitude falls below the long term filter value, so which message is generated? ●
- a. TOO LOW GEAR
 - b. TOO LOW TERRAIN
 - c. DON'T SINK
413. An approach is flown at 140 knots with the gear down and the flaps still up, so which submode watches the terrain clearance? ●
- a. Mode 4B
 - b. Mode 4A
 - c. Mode 4C
414. Which message is common to a mode 4A penetration at 250 knots and to a mode 4C penetration of the filter value ●
- a. TOO LOW TERRAIN
 - b. TOO LOW FLAPS
 - c. TOO LOW GEAR

- 415.** The aircraft is more than two dots below the glide slope at 250 feet radio altitude, so which mode 5 activation has been reached? ●
- The hard alert activation
 - Only the soft alert activation
 - Neither of the two activations
- 416.** The aircraft drifts 1.5 dots below the beam centre at 800 feet radio altitude, so which activation occurs? ●
- The soft alert activation
 - No alert activation at all
 - The hard alert activation
- 417.** A glide slope deviation message is heard noticeably quieter than the other warnings, so what does that tell the crew? ●
- It is the hard alert activation
 - It is the soft alert activation
 - It is a failure of the loudspeaker

Answers — GPWS modes 1 to 5

- 359. c** The EGPWC is the control unit of the system. (L1, PDF p. 907)
- 360. c** Mode 1 is the excessive descent rate mode. (L1, PDF p. 909)
- 361. c** The outer boundary gives the SINK RATE message. (L1, PDF p. 909)
- 362. b** Mode 2 covers an excessive rate of closure with the terrain. (L1, PDF p. 911)
- 363. a** Mode 2B applies for the first 60 seconds after take-off. (L1, PDF p. 911)
- 364. b** Mode 3 is the altitude loss after take-off mode. (L1, PDF p. 913)
- 365. b** Mode 3 gives the DON'T SINK message. (L1, PDF p. 913)
- 366. b** Mode 4 covers unsafe terrain clearance. (L1, PDF p. 915)
- 367. a** Mode 4 has three submodes. (L1, PDF p. 915)
- 368. c** Mode 5 covers a deviation below the glide slope. (L1, PDF p. 917)
- 369. c** There are two levels of mode 5 operation. (L1, PDF p. 917)
- 370. c** Mode 6 offers selectable radio altitude callouts on the landing approach. (L2, PDF p. 907)
- 371. c** The warning function is the one stated to be independent of the others. (L2, PDF p. 907)
- 372. a** The inner boundary gives the PULL UP message. (L2, PDF p. 909)
- 373. c** Those three inputs define the two mode 1 boundaries. (L2, PDF p. 909)
- 374. a** A discrete output drives the visual annunciations in both cases. (L2, PDF p. 909)
- 375. c** Mode 2 alerts are possible in descent, in level flight and in the climb. (L2, PDF p. 911)
- 376. c** Vertical speed and radio altitude are the two signals used. (L2, PDF p. 911)
- 377. b** Mode 2A covers every configuration that is not a mode 2B configuration. (L2, PDF p. 911)
- 378. b** The first mode 2A message is TERRAIN TERRAIN. (L2, PDF p. 911)
- 379. b** PULL UP is repeated until the aircraft exits the warning envelope. (L2, PDF p. 911)
- 380. b** The upper boundary of this mode varies with aircraft speed. (L2, PDF p. 911)
- 381. c** The printed figure is plus or minus 2 dots. (L2, PDF p. 911)
- 382. b** In that configuration the PULL UP messages are suppressed. (L2, PDF p. 911)
- 383. b** The two phases are immediately after take-off and during a missed approach. (L2, PDF p. 913)
- 384. a** They stay active until a positive rate of climb exists. (L2, PDF p. 913)
- 385. a** The alert needs the gear up and the flaps away from the landing position. (L2, PDF p. 913)
- 386. a** Detection is made with reference to the phase of flight and the speed. (L2, PDF p. 915)

- 387. b** Mode 4A covers cruise and approach with the gear not in the landing configuration. (L2, PDF p. 915)
- 388. c** The printed speed for that message is 190 knots or below. (L2, PDF p. 915)
- 389. b** The boundary increases linearly with airspeed. (L2, PDF p. 915)
- 390. c** That message belongs to the gear down, flaps not at landing case. (L2, PDF p. 915)
- 391. a** Mode 4C uses a minimum terrain clearance floor that rises during take-off. (L2, PDF p. 915)
- 392. a** Mode 5 works on a front course ILS approach. (L2, PDF p. 917)
- 393. a** The soft alert is at approximately one half, minus 6dB, of the other levels. (L2, PDF p. 917)
- 394. b** The hard alert belongs to radio altitudes below 300 feet. (L2, PDF p. 917)
- 395. a** The hard alert needs more than 2 dots below the glide slope. (L2, PDF p. 917)
- 396. c** The warning functions keep working as long as their inputs remain available. (L3, PDF p. 907)
- 397. b** The basic GPWS covers modes 1 to 6 while only modes 1 to 5 are the basic requirements, so mode 6 is the one left out. (L3, PDF p. 907)
- 398. a** The warning lamp drivers belong to the alert output processing. (L3, PDF p. 907)
- 399. a** The modes are selected by program pins before the computer is installed. (L3, PDF p. 907)
- 400. a** The first message comes from the outer boundary, so the inner one has still not been crossed. (L3, PDF p. 909)
- 401. b** An excessive descent rate is the trigger of mode 1. (L3, PDF p. 909)
- 402. a** With gear and flaps at landing the repeated message is TERRAIN. (L3, PDF p. 911)
- 403. c** With either gear or flaps not at landing the initial message is TERRAIN TERRAIN. (L3, PDF p. 911)
- 404. c** That configuration is one of those listed for mode 2B. (L3, PDF p. 911)
- 405. c** Mode 1 uses barometric and inertial vertical speed and mode 2 a vertical speed signal, both with radio altitude. (L3, PDF p. 911)
- 406. c** Mode 1 gives PULL UP at the inner boundary and mode 2 repeats PULL UP on continued penetration. (L3, PDF p. 911)
- 407. a** That is exactly the condition printed for the mode 3 alert. (L3, PDF p. 913)
- 408. a** The computer itself calculates the permitted loss before alerting. (L3, PDF p. 913)
- 409. b** Only the mode 3 computation is stated to use altitude above sea level information. (L3, PDF p. 913)
- 410. a** That is the mode 3 case, and mode 4C would need the radio altitude to fall below the filter value. (L3, PDF p. 913)
- 411. a** Above the 190 knot case the penetration of that boundary gives TOO LOW TERRAIN. (L3, PDF p. 915)
- 412. b** That decrease below the filter value gives TOO LOW TERRAIN. (L3, PDF p. 915)
- 413. a** Mode 4B is the submode that applies once the landing gear is lowered. (L3, PDF p. 915)
- 414. a** Mode 4A gives TOO LOW TERRAIN when the boundary reached at 250 knots and above is penetrated, and mode 4C gives it for a fall below the filter value. (L3, PDF p. 915)
- 415. a** Below 300 feet radio altitude and more than 2 dots below the glide slope are the printed conditions of the hard alert activation. (L3, PDF p. 917)
- 416. a** It is beyond 1.3 dots so the soft alert comes, while the hard alert needs below 300 feet and more than 2 dots. (L3, PDF p. 917)
- 417. b** Only the soft alert is given at about half the normal volume. (L3, PDF p. 917)